Well-posedness of fully coupled McKean–Vlasov FBSDEs with jumps under full-tuple law dependence

Chunrong Feng and Christian Garry*

Department of Mathematical Sciences, Durham University

South Road, Durham DH1 3LE, United Kingdom

chunrong.feng@durham.ac.uk

*Corresponding author: christian.t.garry@durham.ac.uk

Abstract

We prove existence, uniqueness, and stability for fully coupled McKean–Vlasov forward–backward SDEs with jumps whose drift, diffusion, jump, and driver coefficients may depend Lipschitz-continuously, in quadratic Wasserstein distance, on the joint law of the full solution tuple $\Theta = (X, Y, Z, U)$: forward state, backward variable, Brownian integrand, and $L^2(\nu)$-valued jump integrand. The terminal function may depend Lipschitz-continuously on X_T and its law. The system is driven by a Brownian motion and an independent compensated Poisson random measure with arbitrary σ-finite intensity, so infinite jump activity is admitted. Both the Lipschitz and monotonicity hypotheses are imposed only along diagonal tuple–law pairs $(\Theta, \mathrm{Law}(\Theta))$; we show that expected diagonal monotonicity is strictly weaker than pointwise monotonicity.

Under a jump-extended G-monotonicity condition we establish an a priori continuous-dependence estimate, uniqueness, and existence on every prescribed finite horizon, by monotone continuation in the coupling strength from a small-coupling base case. A mean-field dealer-market example realises the U-law dependence non-perturbatively: its law interaction is monotone at every interaction strength, and its mark measure has infinite activity.

Keywords: McKean–Vlasov FBSDEs; full-tuple law dependence; infinite jump activity; expected diagonal G-monotonicity; monotone continuation; Wasserstein distance.

MSC 2020: Primary 60H10; secondary 60G57.

1 Introduction

We prove existence, uniqueness, and stability, on every fixed finite horizon T and with no smallness restriction, for the fully coupled McKean–Vlasov forward–backward system with jumps

$$\begin{cases} X_t = \xi_0 + \int_0^t b(s,\Theta_s,\mu_s)\,\mathrm{d}s + \int_0^t \sigma(s,\Theta_s,\mu_s)\,\mathrm{d}W_s + \int_0^t\int_E h(s,\Theta_s^-,e,\mu_s^-)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e), \\ Y_t = g(X_T,\mathrm{Law}(X_T)) + \int_t^T f(s,\Theta_s,\mu_s)\,\mathrm{d}s - \int_t^T Z_s\,\mathrm{d}W_s - \int_t^T\int_E U_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e), \end{cases} \tag{1.1}$$

driven by a Brownian motion W and an independent compensated Poisson random measure $\widetilde{N}$. The intensity ν is assumed only σ-finite: infinite jump activity, $\nu(E)=\infty$, is admitted. The state space is the Hilbert space

$$\mathcal{H} := \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \times L^2(\nu;\mathbb{R}^m),$$

carrying the norm

$$|\theta|_{\mathcal{H}}^2 := |x|^2 + |y|^2 + |z|_F^2 + \|u\|_{L^2(\nu)}^2, \qquad \theta = (x,y,z,u),$$

where $|\cdot|$ is the Euclidean norm (on any $\mathbb{R}^j$) and the Frobenius and $L^2(\nu;\mathbb{R}^m)$ norms are

$$|z|_F^2 := \sum_{i,j} z_{ij}^2, \qquad \|u\|_{L^2(\nu)}^2 := \int_E |u(e)|^2\,\nu(\mathrm{d}e);$$

$L^2(\nu;\mathbb{R}^m)$ is henceforth abbreviated $L^2(\nu)$.

We seek a solution tuple

$$\Theta_t = (X_t, Y_t, Z_t, U_t);$$

the jump coefficient is evaluated at the predictable version $\Theta_t^- := (X_{t-}, Y_{t-}, Z_t, U_t)$, which takes left limits in the forward and backward states (Z, U are already predictable). The coefficients are evaluated along the joint-law flows

$$\mu_t = \mathrm{Law}(\Theta_t), \qquad \mu_t^- := \mathrm{Law}(\Theta_t^-) \in \mathcal{P}_2(\mathcal{H}).$$

The distinguishing feature is the law argument: the coefficients are Lipschitz in the quadratic Wasserstein distance W_2 on $\mathcal{P}_2(\mathcal{H})$, so the law of the $L^2(\nu)$-valued jump integrand U itself may enter the coefficients; both the Lipschitz and the monotonicity hypotheses are imposed only along the solution's own tuple–law pairs $(\Theta, \mathrm{Law}(\Theta))$.

This class of equations is motivated by the optimality systems of mean-field games and mean-field control problems with jumps, where the law argument is typically the state or control law, and by their numerical analysis: the a posteriori error estimates of Reisinger, Stockinger and Zhang [19] for Brownian MV-FBSDEs rest on the Brownian case of the monotonicity hypothesis used below, and carrying them to jumps needs the well-posedness theory proved here.

The primary comparison is the theorem of Li and Min [14]: global well-posedness for fully coupled mean-field FBSDEs with jumps, proved by continuation under a jump-extended G-monotonicity condition [14, (H3.2)]. Their mean-field interaction is of expectation-functional (independent-copy) type, their jump integrand enters only through its pointwise values, and their monotonicity hypothesis imposes finite total jump mass; the present theorem removes all three restrictions, allowing W_2-dependence on the full tuple law, the law of U included, at arbitrary σ-finite activity, under the expected G-monotonicity condition set out below. Example 5.3 details the embedding of their interaction class.

For Brownian noise, well-posedness of fully coupled mean-field FBSDEs under joint-law dependence has been studied by Bensoussan, Yam and Zhang [3], Chen and Nie [6] and others [4, 7, 10, 17, 21, 23, 26] under various monotonicity and weak-coupling assumptions. All of these are Brownian: no jump channel, no integrand U.

With jumps but no mean-field coupling, fully coupled FBSDEs are well posed over arbitrary horizons by the same monotone-continuation method: Wu [24] for the Brownian–Poisson case, and domination-monotonicity continuation carried to the Lévy-driven case by Xu, Tang and Meng [25]. For jump *mean-field* systems in decoupled form, the well-posedness and regularity theory is due to Li [13]. Neither setting addresses the fully coupled problem with full-tuple law dependence.

Further fully coupled jump mean-field results trade scope for other restrictions. Matoussi, Manai and Salhi [16] treat W_2-dependence on the (X, Y)-marginal law at finite activity, under monotonicity hypotheses and explicit size restrictions on the Lipschitz constants. Chen, Shi and Wu [5] work on an arbitrary fixed horizon in the weakly coupled regime, where quantitative weak coupling replaces the structural dissipativity used here. Liu and Zhang [15] allow pointwise dependence on Y, Z but take the measure argument to be $\mathrm{Law}(X)$, on a small time interval and under finite jump activity. In a control direction, mean-field-type control driven by jump-diffusions is solved globally in time by Bensoussan, Huang, Tang and Yam [2], with state-law dependence and finite total jump mass.

Assumption 2.5 below adjoins a jump term to the monotonicity hypothesis of [19, (H.1)(1)], posed there, as in the condition [3, (A1)], on the joint law of (X, Y, Z) with a rectangular full-rank G. The measure argument is correspondingly the joint law of the full tuple (X, Y, Z, U), so the law of the $L^2(\nu)$-valued jump integrand may enter the coefficients, and ν is σ-finite, infinite activity included.

The works surveyed above are each restricted on at least one of the three axes (measure argument, jump activity, coupling strength). We prove well-posedness under the stated L^2 and monotonicity assumptions by global-in-time monotone continuation; concretely:

(i) an a priori stability estimate: the full difference of two solutions is controlled by the differences of their coefficients and data (Theorem 3.1);

(ii) uniqueness in the natural class (Theorem 3.3);

(iii) global existence by monotone continuation from a small-coupling base case (Theorem 3.5);

(iv) robustness: well-posedness survives full-tuple-law perturbations below the dissipativity margin at equal forward and backward dimension (Corollary 3.6; Remark 3.7 for general rank); and

(v) strictness: expected diagonal monotonicity is strictly weaker than its pointwise form (Proposition 3.8).

The paper is organised as follows. Section 2 sets out the model and its assumptions. Section 3 states the main results. Section 4 carries out the continuation proof. Section 5 develops the examples. Section 6 concludes and identifies open extensions. Appendix A constructs the Borel law flow used throughout.

2 The Brownian–Poisson MV-FBSDEJ

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be a filtered probability space satisfying the usual conditions, over a fixed horizon $T > 0$ and dimensions $n, m, d \geq 1$, carrying a d-dimensional Brownian motion W and an independent Poisson random measure N on $\mathbb{R}_+ \times E$ with intensity $\mathrm{d}t\, \nu(\mathrm{d}e)$, where ν is a σ-finite measure on a countably generated $(E, \mathcal{E})$ (so that $L^2(\nu; \mathbb{R}^j)$ is separable for every $j \geq 1$). The filtration $(\mathcal{F}_t)$ is the usual augmentation of the one generated by W, N and an independent initial σ-field $\mathcal{F}_0$. We do not require ν to be finite: the infinite-activity regime, in which N has infinitely many atoms on every nondegenerate time interval, is admitted throughout, with the finite-activity case $\nu(E) < \infty$ recovered as a special instance.

Write $\widetilde{N}(\mathrm{d}t, \mathrm{d}e) := N(\mathrm{d}t, \mathrm{d}e) - \mathrm{d}t\, \nu(\mathrm{d}e)$ for the associated compensated measure. For predictable $\psi \geq 0$ the compensation formula reads

$$\mathbb{E}\int_0^T \int_E \psi_s(e)\, N(\mathrm{d}s, \mathrm{d}e) = \mathbb{E}\int_0^T \int_E \psi_s(e)\, \nu(\mathrm{d}e)\, \mathrm{d}s \in [0, \infty], \tag{2.1}$$

and the zero-mean property of the compensated integral reads

$$\mathbb{E}\int_0^T \int_E \psi_s(e)\, \widetilde{N}(\mathrm{d}s, \mathrm{d}e) = 0 \qquad \text{whenever} \quad \mathbb{E}\int_0^T \int_E |\psi_s(e)|\, \nu(\mathrm{d}e)\, \mathrm{d}s < \infty. \tag{2.2}$$

The unknown is the tuple $\Theta = (X, Y, Z, U)$, with law $\mu_t = \mathrm{Law}(\Theta_t)$ and predictable counterparts Θ_t^-, μ_t^- introduced below; the continuous MV-FBSDEJ is the system (1.1) displayed in the introduction. The tuple $\Theta_t = (X_t, Y_t, Z_t, U_t)$ takes values in the Hilbert space $\mathcal{H}$, componentwise

$$X_t \in \mathbb{R}^n, \qquad Y_t \in \mathbb{R}^m, \qquad Z_t \in \mathbb{R}^{m\times d}, \qquad U_t \in L^2(\nu;\mathbb{R}^m).$$

The jump terms are evaluated at the predictable tuple

$$\Theta_t^- = (X_{t-}, Y_{t-}, Z_t, U_t), \qquad \mu_t^- = \mathrm{Law}(\Theta_t^-),$$

in which the forward and backward states are replaced by their left limits while the integrands Z_t, U_t, already predictable, are left untouched. Throughout, μ_t and μ_t^- denote the Borel representatives fixed in Lemma A.1, which agree with $\mathrm{Law}(\Theta_t)$ and $\mathrm{Law}(\Theta_t^-)$ for a.e. t. Expressions such as $b(t, \Theta_t, \mu_t)$ are thereby defined at every t, while every claim made about them is $\mathrm{d}t$-a.e. or time-integrated.

The filtration generated by W and N has the *Brownian–Poisson predictable representation property* ([1, §5.3]; [9, Theorem 13.49]), and it passes to the independent initial enlargement by $\mathcal{F}_0$ in the standard way: every square-integrable $(\mathcal{F}_t)$-martingale M can be written as

$$M_t = M_0 + \int_0^t Z_s \,\mathrm{d}W_s + \int_0^t\int_E U_s(e)\,\widetilde{N}(\mathrm{d}s, \mathrm{d}e), \qquad \text{for every } t \in [0,T],$$

for an $\mathcal{F}_0$-measurable M_0 and a predictable square-integrable pair (Z, U), and this pair is unique up to null sets.

Definition 2.1 (Diagonal input)**.** Fix $t \in [0,T]$. A *diagonal input at time* t is a pair (Θ, μ) consisting of a square-integrable $\mathcal{F}_t$-measurable tuple $\Theta \in L^2(\Omega, \mathcal{F}_t; \mathcal{H})$, on the standing basis, *together with its own law* $\mu = \mathrm{Law}(\Theta) \in \mathcal{P}_2(\mathcal{H})$; such pairs form the diagonal

$$\mathcal{D}_t := \Big\{ \big(\Theta, \mathrm{Law}(\Theta)\big) : \Theta \in L^2(\Omega, \mathcal{F}_t; \mathcal{H}) \Big\} \;\subset\; L^2(\Omega, \mathcal{F}_t; \mathcal{H}) \times \mathcal{P}_2(\mathcal{H}).$$

All hypotheses below are imposed along $\mathcal{D}_t$ only: the measure argument is never varied independently of the tuple. Throughout, a lower-case θ denotes a *point* of $\mathcal{H}$ and an upper-case Θ denotes a *random* $\mathcal{H}$-valued tuple.

Difference notation. For two diagonal inputs $(\Theta, \mu), (\Theta', \mu')$ at the same time t on the same probability space, δ denotes the difference

$$\delta X := X - X', \qquad \delta\psi := \psi(t, \Theta, \mu) - \psi(t, \Theta', \mu'), \quad \psi \in \{b, \sigma, h, f\},$$

and, with the *induced* terminal laws $\rho := \mathrm{Law}(X_T)$ and $\rho' := \mathrm{Law}(X'_T)$,

$$\delta g_T := g(X_T, \rho) - g(X'_T, \rho').$$

Throughout, b, σ, h, f, g are deterministic functions of their arguments; randomness enters only through the inputs and, in Section 4, through additive square-integrable forcings (the ω-random coefficients of [6, 14] are not pursued here). Their signatures are

$$
\begin{aligned}
b\colon\ & [0,T]\times\mathcal{H}\times\mathcal{P}_2(\mathcal{H}) \ \to\ \mathbb{R}^n,\\
\sigma\colon\ & [0,T]\times\mathcal{H}\times\mathcal{P}_2(\mathcal{H}) \ \to\ \mathbb{R}^{n\times d},\\
h\colon\ & [0,T]\times\mathcal{H}\times\mathcal{P}_2(\mathcal{H}) \ \to\ L^2(\nu;\mathbb{R}^n),\\
f\colon\ & [0,T]\times\mathcal{H}\times\mathcal{P}_2(\mathcal{H}) \ \to\ \mathbb{R}^m,\\
g\colon\ & \mathbb{R}^n\times\mathcal{P}_2(\mathbb{R}^n) \ \to\ \mathbb{R}^m,
\end{aligned}
$$

the terminal function alone carrying no time argument and depending on the state and its law. All coefficients are jointly Borel measurable in their arguments; for the jump coefficient, whose values lie in the fibre, this is taken as Borel measurability of the map

$$
(t,\theta,\mu) \ \longmapsto\ h(t,\theta,\cdot,\mu), \qquad [0,T]\times\mathcal{H}\times\mathcal{P}_2(\mathcal{H}) \ \longrightarrow\ L^2(\nu;\mathbb{R}^n). \tag{2.3}
$$

Since $L^2(\nu;\mathbb{R}^n)$ is separable, (2.3) follows from joint pointwise measurability of $(t,\theta,e,\mu)\mapsto h(t,\theta,e,\mu)$: the latter makes h weakly measurable by Tonelli, and the Pettis measurability theorem [11, Theorem 1.1.6] then upgrades weak to strong measurability.

For predictable processes we use the Fubini isometry

$$
\begin{gathered}
\mathcal{P} := \text{predictable } \sigma\text{-field on } \Omega\times[0,T] \text{ under } \mathrm{d}t\otimes\mathrm{d}\mathbb{P},\\
L^2(\mathcal{P};L^2(\nu)) \ \cong\ L^2(\mathcal{P}\otimes\mathcal{E})
\end{gathered} \tag{2.4}
$$

of [11, Propositions 1.2.24 and 1.2.25], which supplies a $\mathcal{P}\otimes\mathcal{E}$-measurable representative, unique up to $\mathrm{d}t\otimes\mathrm{d}\mathbb{P}\otimes\nu$-null sets; every Poisson integral below, and every square-integrable $L^2(\nu)$-valued predictable integrand appearing in one, is read through such a representative. In particular, a pointwise formula $u(e)$ for $u\in L^2(\nu)$ is meaningful only when it descends to a well-defined element of $L^2(\nu)$, independent of the representative of u. Consequently, along any diagonal input,

$$
\begin{aligned}
(s,\omega)&\mapsto\psi\Big(s,\Theta_s(\omega),\mu_s\Big), \quad \psi\in\{b,\sigma,f\}, && \text{progressively measurable},\\
(s,\omega,e)&\mapsto h\Big(s,\Theta_{s-}(\omega),e,\mu_s^-\Big), && \text{predictable}.
\end{aligned} \tag{2.5}
$$

The well-definedness of the law flows $t\mapsto\mu_t,\mu_t^-$ and the independence of all time-integrated coefficient identities from the choice of predictable representatives of Z,U are established in Lemma A.1 in Appendix A.

Assumption 2.2 (Integrability at the origin)**.** $\xi_0\in L^2(\Omega,\mathcal{F}_0;\mathbb{R}^n)$ and, with δ_0 the Dirac mass at the relevant origin, the origin values satisfy

$$
\begin{aligned}
b(\cdot,0,\delta_0) \ &\in\ L^2(0,T;\mathbb{R}^n), & \sigma(\cdot,0,\delta_0) \ &\in\ L^2(0,T;\mathbb{R}^{n\times d}),\\
h(\cdot,0,\cdot,\delta_0) \ &\in\ L^2\Big(0,T;L^2(\nu;\mathbb{R}^n)\Big), & f(\cdot,0,\delta_0) \ &\in\ L^2(0,T;\mathbb{R}^m).
\end{aligned}
$$

Assumption 2.3 (Lipschitz coefficients)**.** There is $L \geq 0$ such that, for a.e. $t \in [0, T]$ and all $(\Theta, \mu), (\Theta', \mu') \in \mathcal{D}_t$ (Definition 2.1),

$$|\delta b| + |\delta\sigma|_F + \|\delta h\|_{L^2(\nu)} + |\delta f| \;\leq\; L\Big(|\delta X| + |\delta Y| + |\delta Z|_F + \|\delta U\|_{L^2(\nu)} + W_2(\mu, \mu')\Big)$$

almost surely; and, for all $X_T, X_T' \in L^2(\Omega, \mathcal{F}_T; \mathbb{R}^n)$,

$$|\delta g_T| \;\leq\; L\Big(|X_T - X_T'| + W_2(\rho, \rho')\Big) \tag{2.6}$$

almost surely.

Assumption 2.4 (Coupling matrix)**.** $G \in \mathbb{R}^{m\times n}$ has full rank. Write $\|G\|$ for its operator norm, so that $|Gv| \leq \|G\|\,|v|$ and $|G^\top v| \leq \|G\|\,|v|$, and $s_{\min}(G) > 0$ for its smallest singular value; set

$$c_G := s_{\min}(G)^{-2}.$$

Then

$$|x|^2 \leq c_G|Gx|^2 \quad (m \geq n), \qquad\qquad |y|^2 \leq c_G|G^\top y|^2 \quad (n \geq m).$$

When $G^\top$ is injective the second bound extends to the matrix- and $L^2(\nu)$-valued arguments,

$$|Z|_F^2 \leq c_G|G^\top Z|_F^2, \qquad\qquad \|U\|^2_{L^2(\nu)} \leq c_G\|G^\top U\|^2_{L^2(\nu)}.$$

Assumption 2.5 (Terminal and interior jump-extended G-monotonicity)**.** For all $X_T, X_T' \in L^2(\Omega, \mathcal{F}_T; \mathbb{R}^n)$,

$$\mathbb{E}\langle \delta g_T,\, G\,\delta X_T\rangle \geq \alpha\,\mathbb{E}|G\delta X_T|^2,$$

and, for a.e. $t \in [0, T]$ and all $(\Theta, \mu), (\Theta', \mu') \in \mathcal{D}_t$,

$$\begin{aligned}\mathbb{E}\Big[\langle \delta b,\, G^\top\delta Y\rangle + \langle \delta\sigma,\, G^\top\delta Z\rangle_F + \langle -\delta f,\, G\delta X\rangle + \int_E \langle \delta h(e),\, G^\top\delta U(e)\rangle \nu(\mathrm{d}e)\Big]& \\ \leq -\beta_1\mathbb{E}\Big[|G^\top\delta Y|^2 + |G^\top\delta Z|_F^2 + \|G^\top\delta U\|^2_{L^2(\nu)}\Big] - \beta_2\mathbb{E}|G\delta X|^2&\end{aligned} \tag{2.7}$$

The constants satisfy

$$\alpha, \beta_1, \beta_2 \geq 0, \qquad \alpha + \beta_1 > 0, \qquad \beta_1 + \beta_2 > 0,$$

with $\beta_1 > 0$ if $m < n$ and $\alpha, \beta_2 > 0$ if $m > n$.

Remark 2.6 (Wasserstein bounds). The pair (Θ, Θ') realises a coupling of (μ, μ') on the common probability space, so the W_2-infimum [22, Definition 6.1] is bounded by its synchronous transport cost:

$$W_2(\mu, \mu')^2 \leq \mathbb{E}\Big[|\delta X|^2 + |\delta Y|^2 + |\delta Z|_F^2 + \|\delta U\|^2_{L^2(\nu)}\Big], \tag{2.8}$$

$$W_2(\rho, \rho')^2 \leq \mathbb{E}|\delta X|^2, \tag{2.9}$$

(2.9) being the X-marginal case of (2.8).

In the other direction, let K be a separable Hilbert space and $\bar\phi : \mathcal{H} \to K$ Lipschitz. The law functional $\mu \mapsto \int_{\mathcal{H}} \bar\phi \, \mathrm{d}\mu$ is then W_2-Lipschitz with the same constant: writing the difference as an integral against an optimal coupling π of (μ, μ'), the Lipschitz bound and Jensen give

$$\begin{aligned}\left| \int_{\mathcal{H}} \bar\phi \, \mathrm{d}\mu - \int_{\mathcal{H}} \bar\phi \, \mathrm{d}\mu' \right|_K &= \left| \int_{\mathcal{H}\times\mathcal{H}} \big(\bar\phi(\theta) - \bar\phi(\theta')\big) \, \mathrm{d}\pi \right|_K \\ &\le \mathrm{Lip}(\bar\phi) \int_{\mathcal{H}\times\mathcal{H}} |\theta - \theta'|_{\mathcal{H}} \, \mathrm{d}\pi \\ &\le \mathrm{Lip}(\bar\phi) \Big(\int_{\mathcal{H}\times\mathcal{H}} |\theta - \theta'|_{\mathcal{H}}^2 \, \mathrm{d}\pi \Big)^{1/2} \\ &\le \mathrm{Lip}(\bar\phi) \, W_2(\mu, \mu'). \end{aligned} \tag{2.10}$$

Marginal projection $\mathcal{P}_2(\mathcal{H}) \to \mathcal{P}_2(L^2(\nu))$ is 1-Lipschitz for W_2 by the same argument: write $\mu^U \in \mathcal{P}_2(L^2(\nu))$ for the U-marginal of μ. The image of this optimal π under $(\theta, \theta') \mapsto (u, u')$ is a coupling of (μ^U, μ'^U), so bounding $W_2(\mu^U, \mu'^U)$ by its cost gives

$$\begin{aligned} W_2(\mu^U, \mu'^U)^2 &\le \int_{\mathcal{H}\times\mathcal{H}} \|u - u'\|_{L^2(\nu)}^2 \, \mathrm{d}\pi \\ &\le \int_{\mathcal{H}\times\mathcal{H}} |\theta - \theta'|_{\mathcal{H}}^2 \, \mathrm{d}\pi \\ &\le W_2(\mu, \mu')^2. \end{aligned} \tag{2.11}$$

Remark 2.7 (The coupling matrix G). The full-rank matrix $G \in \mathbb{R}^{m\times n}$ serves to pair the forward and backward components when $n \neq m$; its full-rank property ensures the resulting G-coercivity controls the full forward variables when $m \ge n$, or the full backward variables when $n \ge m$.

Remark 2.8 (Expected diagonal monotonicity). Assumption 2.5 is imposed *in expectation*. The *pointwise* G-monotonicity condition requires the inequality of (2.7), without expectations, at every fixed $\theta, \theta' \in \mathcal{H}$ and $\mu, \mu' \in \mathcal{P}_2(\mathcal{H})$. Applying it at

$$\theta = \Theta(\omega), \qquad \theta' = \Theta'(\omega), \qquad \mu = \mathrm{Law}(\Theta), \qquad \mu' = \mathrm{Law}(\Theta')$$

and taking expectations gives Assumption 2.5, so the pointwise condition is the stronger one. The converse fails: the expectation may average sign-indefinite pointwise pairings into a dissipative one (Proposition 3.8). Example 5.1(i) verifies Assumption 2.5 exactly, Example 5.4 perturbatively.

Remark 2.9 (Coupling formulation). For $t > 0$ the σ-field $\mathcal{F}_t$ supports a uniformly distributed variable (a function of W_t^1), so every coupling of two laws in $\mathcal{P}_2(\mathcal{H})$ is the joint law of a pair of $\mathcal{F}_t$-measurable inputs, by the kernel representation lemma [12, Lemma 4.22]. The diagonal hypotheses, Assumptions 2.3 and 2.5, are therefore intrinsic: properties of the coefficients, not of the standing basis. As the interior conditions are imposed $\mathrm{d}t$-a.e. and the terminal ones at $T > 0$, no richness assumption on $\mathcal{F}_0$ is needed. The proofs use only the expected diagonal form.

Remark 2.10 (The β_1 dichotomy in the jump channel). Taking $\delta X = \delta Y = \delta Z = 0$ and δU arbitrary in (2.7) reduces the interior pairing to

$$\mathbb{E}\int_E \langle \delta h(e),\, G^\top \delta U(e)\rangle\, \nu(\mathrm{d}e) \le -\beta_1\, \mathbb{E}\|G^\top \delta U\|^2_{L^2(\nu)},$$

so $\beta_1 > 0$ forces the jump coefficient h to be strictly U-dissipative. A jump coefficient of the natural additive form $h = h(t, x, e)$ carries no U-feedback, so $\delta h = 0$ while δU may be chosen with $G^\top \delta U \neq 0$; the displayed bound then forces $\beta_1 \le 0$, hence $\beta_1 = 0$ as the constants are non-negative. Assumption 2.5 then requires $m \ge n$ with $\alpha, \beta_2 > 0$, the G-injective branch, and the $m < n$ rank case is available only for U-dissipative jump coefficients. The restriction is structural: it follows from Assumption 2.5 itself, not from the estimates that exploit it. Recovering the additive form at $m < n$ under a different structural condition remains open.

Remark 2.11 (Dissipativity margin). Say that (2.7) holds *with margin* $c > 0$ if

$$\begin{aligned}
&\mathbb{E}\Big[\langle \delta b,\, G^\top \delta Y\rangle + \langle \delta\sigma,\, G^\top \delta Z\rangle_F + \langle -\delta f,\, G\delta X\rangle + \int_E \langle \delta h(e),\, G^\top \delta U(e)\rangle \nu(\mathrm{d}e)\Big] \\
&\quad \le -\beta_1 \mathbb{E}\Big[|G^\top \delta Y|^2 + |G^\top \delta Z|_F^2 + \|G^\top \delta U\|^2_{L^2(\nu)}\Big] - \beta_2 \mathbb{E}|G\delta X|^2 \\
&\qquad - c\,\mathbb{E}\Big[|G^\top \delta Y|^2 + |G^\top \delta Z|_F^2 + \|G^\top \delta U\|^2_{L^2(\nu)} + |G\delta X|^2\Big].
\end{aligned} \tag{2.12}$$

Suppose the interior pairing has been bounded with constants $\bar\beta_1, \bar\beta_2$. For every $0 < c < \min\{\bar\beta_1, \bar\beta_2\}$ the same bound is (2.12) with the reduced constants $\beta_i := \bar\beta_i - c$ and margin c: recording weaker constants turns the difference into slack, which a perturbation raising the interior pairing by at most the margin leaves intact. When $m = n$ the margin may instead be recorded in the *unweighted* form

$$\begin{aligned}
&\mathbb{E}\Big[\langle \delta b,\, G^\top \delta Y\rangle + \langle \delta\sigma,\, G^\top \delta Z\rangle_F + \langle -\delta f,\, G\delta X\rangle + \int_E \langle \delta h(e),\, G^\top \delta U(e)\rangle \nu(\mathrm{d}e)\Big] \\
&\quad \le -\beta_1 \mathbb{E}\Big[|G^\top \delta Y|^2 + |G^\top \delta Z|_F^2 + \|G^\top \delta U\|^2_{L^2(\nu)}\Big] - \beta_2 \mathbb{E}|G\delta X|^2 \\
&\qquad - c\,\mathbb{E}\Big[|\delta Y|^2 + |\delta Z|_F^2 + \|\delta U\|^2_{L^2(\nu)} + |\delta X|^2\Big].
\end{aligned} \tag{2.13}$$

3 Main results

Solution space. Write

$$\begin{aligned}
L^2(0,T;\mathbb{R}^j) &:= \Big\{V \in \mathbb{R}^j \text{ càdlàg adapted } : \mathbb{E}\int_0^T |V_t|^2\, \mathrm{d}t < \infty\Big\}, \\
\mathcal{S}^2(0,T;\mathbb{R}^j) &:= \Big\{V \in \mathbb{R}^j \text{ càdlàg adapted } : \mathbb{E}\sup_{0\le t\le T} |V_t|^2 < \infty\Big\}, \\
\mathcal{H}^2_W(0,T;\mathbb{R}^{m\times d}) &:= \Big\{Z \in \mathbb{R}^{m\times d} \text{ predictable } : \mathbb{E}\int_0^T |Z_t|_F^2\, \mathrm{d}t < \infty\Big\}, \\
\mathcal{H}^2_N(0,T;L^2(\nu;\mathbb{R}^m)) &:= \Big\{U \in L^2(\nu;\mathbb{R}^m) \text{ predictable } : \mathbb{E}\int_0^T \|U_t\|^2_{L^2(\nu)}\, \mathrm{d}t < \infty\Big\}.
\end{aligned}$$

Set

$$\mathbb{S} := \mathcal{S}^2(0,T;\mathbb{R}^n) \times \mathcal{S}^2(0,T;\mathbb{R}^m) \times \mathcal{H}^2_W(0,T;\mathbb{R}^{m\times d}) \times \mathcal{H}^2_N(0,T;L^2(\nu;\mathbb{R}^m)),$$
$$\mathbb{H}^2 := L^2(0,T;\mathbb{R}^n) \times L^2(0,T;\mathbb{R}^m) \times \mathcal{H}^2_W(0,T;\mathbb{R}^{m\times d}) \times \mathcal{H}^2_N(0,T;L^2(\nu;\mathbb{R}^m)),$$

with squared norm

$$\|\Theta\|^2_{\mathbb{S}} := \mathbb{E}\sup_{t\le T}|X_t|^2 + \mathbb{E}\sup_{t\le T}|Y_t|^2 + \mathbb{E}\int_0^T \Big(|Z_t|_F^2 + \|U_t\|^2_{L^2(\nu)}\Big)\mathrm{d}t.$$

The elements are equivalence classes: càdlàg components are identified when indistinguishable, Z-components when they agree $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$-a.e., and U-components when they agree $\mathrm{d}t \otimes \mathrm{d}\mathbb{P} \otimes \nu$-a.e. Under this identification $\|\cdot\|_{\mathbb{S}}$ is a norm rather than a seminorm, and $(\mathbb{S}, \|\cdot\|_{\mathbb{S}})$ is a Banach space. Since $\mathbb{E}\int_0^T |V_t|^2\,\mathrm{d}t \le T\,\mathbb{E}\sup_{t\le T}|V_t|^2$, the inclusion $\mathbb{S} \subset \mathbb{H}^2$ holds.

A *solution* of (1.1) is a tuple $\Theta = (X, Y, Z, U) \in \mathbb{S}$ satisfying both identities of (1.1) almost surely for every $t \in [0, T]$, with h evaluated at the predictable tuple (Θ_t^-, μ_t^-). Both sides of each identity are càdlàg in t, so the exceptional null set may be taken independent of t.

The stability and uniqueness theorems below rest on comparing solutions of (1.1) under two coefficient systems, indexed by $k = 1, 2$. We write $\delta\cdot = \cdot^1 - \cdot^2$, with $\mu_t^k := \mathrm{Law}(\Theta_t^k)$, and evaluate every coefficient difference along solution 2:

$$\begin{aligned}
(\delta\psi)(t) &:= \psi^1(t, \Theta_t^2, \mu_t^2) - \psi^2(t, \Theta_t^2, \mu_t^2), \qquad\qquad \psi \in \{b, \sigma, f\},\\
(\delta h)(t) &:= h^1(t, \Theta_t^{2,-}, \cdot, \mu_t^{2,-}) - h^2(t, \Theta_t^{2,-}, \cdot, \mu_t^{2,-}),\\
\widehat{\delta g}_T &:= g^1\Big(X_T^2, \mathrm{Law}(X_T^2)\Big) - g^2\Big(X_T^2, \mathrm{Law}(X_T^2)\Big).
\end{aligned}$$

Theorem 3.1 (Stability under coefficient and data perturbation)**.** *Let the coefficient system $(b^1, \sigma^1, h^1, f^1, g^1)$, with initial datum ξ_0^1, satisfy Assumptions 2.2–2.5 with constants $L, \alpha, \beta_1, \beta_2$ and coupling matrix G; let $(b^2, \sigma^2, h^2, f^2, g^2)$, with initial datum ξ_0^2, satisfy Assumptions 2.2–2.3 with its own Lipschitz constant; and let $\Theta^k \in \mathbb{S}$ solve (1.1) for system k. Then*

$$\begin{aligned}
&\mathbb{E}\sup_{t\le T}|\delta X_t|^2 + \mathbb{E}\sup_{t\le T}|\delta Y_t|^2 + \mathbb{E}\int_0^T \Big(|\delta Z_t|_F^2 + \|\delta U_t\|^2_{L^2(\nu)}\Big)\mathrm{d}t\\
&\quad\le C_{\mathrm{stab}}\bigg(\mathbb{E}|\delta\xi_0|^2 + \mathbb{E}\Big|\widehat{\delta g}_T\Big|^2\\
&\qquad\qquad + \mathbb{E}\int_0^T \Big(|(\delta b)(t)|^2 + |(\delta\sigma)(t)|_F^2 + \|(\delta h)(t)\|^2_{L^2(\nu)} + |(\delta f)(t)|^2\Big)\mathrm{d}t\bigg),
\end{aligned} \tag{3.1}$$

where C_{stab} depends only on $T, L, \|G\|, c_G, \alpha, \beta_1, \beta_2$ and the universal Burkholder–Davis–Gundy (BDG) constant c_{BDG}. The dependence is on system 1 alone: system 2 enters (3.1) only through its right-hand side. Neither the dimensions n, m, d nor the total mass $\nu(E)$ appears, so the estimate is uniform over the infinite-activity regime.

The proof, a reduction to Lemma 4.3 at $\lambda = 1$, is in Section 4.4.

If system 2 also satisfies Assumptions 2.4–2.5, with its own coupling matrix and constants, exchanging the roles gives the companion bound with the differences evaluated along solution 1 and the constant built from system 2's data. When the two systems share the four interior coefficients, the bound gives continuous dependence on the initial and terminal data. More minimally, the hypotheses on system 2 serve only to make the right-hand side of (3.1) finite: the estimate holds whenever the displayed differences are square-integrable.

Remark 3.2 (Solution class). Let $\Theta = (X, Y, Z, U) \in \mathbb{H}^2$ satisfy (1.1). Then $\Theta \in \mathbb{S}$. Indeed, the coefficients evaluated along Θ lie in $L^2(\mathrm{d}t \otimes \mathrm{d}\mathbb{P})$ by Assumptions 2.2–2.3, so Doob/BDG applied to the forward equation gives

$$\mathbb{E} \sup_{t \le T} |X_t|^2 < \infty,$$

whence $g(X_T, \mathrm{Law}(X_T)) \in L^2$ by Assumption 2.3; taking $t = 0$ in the backward equation gives $Y_0 \in L^2$. Subtracting that identity from the backward equation puts Y in forward form,

$$Y_t = Y_0 - \int_0^t f(s, \Theta_s, \mu_s)\,\mathrm{d}s + \int_0^t Z_s\,\mathrm{d}W_s + \int_0^t\!\int_E U_s(e)\,\widetilde{N}(\mathrm{d}s, \mathrm{d}e),$$

so Cauchy–Schwarz on the drift and Doob/BDG on the two martingales give

$$\mathbb{E} \sup_{t \le T} |Y_t|^2 < \infty.$$

Theorem 3.3 (Uniqueness). *Under Assumptions 2.3–2.5, the system* (1.1) *has at most one solution. More precisely, if $\Theta^1, \Theta^2 \in \mathbb{H}^2$ satisfy* (1.1) *with the same initial condition, then $\Theta^1 = \Theta^2$. Furthermore, $\Theta^1 - \Theta^2 \in \mathbb{S}$.*

Proof. Let $\Theta^k = (X^k, Y^k, Z^k, U^k)$, $k = 1, 2$, be two such tuples, and write $\delta\Theta_t := \Theta^1_t - \Theta^2_t$, $\mu^k_t := \mathrm{Law}(\Theta^k_t)$ and, for $\psi \in \{b, \sigma, h, f\}$,

$$\delta\psi_t := \psi(t, \Theta^1_t, \mu^1_t) - \psi(t, \Theta^2_t, \mu^2_t).$$

By assumption

$$\mathbb{E}\int_0^T \Big(|X^k_t|^2 + |Y^k_t|^2 + |Z^k_t|_F^2 + \|U^k_t\|^2_{L^2(\nu)}\Big)\mathrm{d}t < \infty, \qquad k = 1, 2,$$

so by Fubini

$$\mathbb{E}|X^k_t|^2 + \mathbb{E}|Y^k_t|^2 + \mathbb{E}|Z^k_t|_F^2 + \mathbb{E}\|U^k_t\|^2_{L^2(\nu)} < \infty \qquad \text{for a.e. } t \in [0, T],$$

that is, $\Theta^k_t \in L^2(\Omega, \mathcal{F}_t; \mathcal{H})$ for a.e. t. Thus $(\Theta^k_t, \mu^k_t) \in \mathcal{D}_t$ for a.e. t, and Assumption 2.3 is applicable to the pair (Θ^1_t, μ^1_t), (Θ^2_t, μ^2_t); moreover $W_2(\mu^1_t, \mu^2_t)^2 \le \mathbb{E}|\delta\Theta_t|^2_{\mathcal{H}}$ by (2.8). Hence

Assumption 2.3 yields

$$\mathbb{E}\Big[|\delta b_t|^2 + |\delta\sigma_t|_F^2 + \|\delta h_t\|^2_{L^2(\nu)} + |\delta f_t|^2\Big] \le C\,\mathbb{E}|\delta\Theta_t|^2_{\mathcal{H}}$$

for a.e. t, where C is independent of t. Then

$$\mathbb{E}\int_0^T \Big[|\delta b_t|^2 + |\delta\sigma_t|_F^2 + \|\delta h_t\|^2_{L^2(\nu)} + |\delta f_t|^2\Big]\mathrm{d}t < \infty. \tag{3.2}$$

We next show that $\Theta^1 - \Theta^2 \in \mathbb{S}$. Since the two solutions have the same initial condition, the forward equation is

$$\delta X_t = \int_0^t \delta b_s\,\mathrm{d}s + \int_0^t \delta\sigma_s\,\mathrm{d}W_s + \int_0^t\int_E \delta h_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e).$$

By Cauchy–Schwarz, the BDG inequality and the compensation formula,

$$\mathbb{E}\sup_{0\le t\le T}|\delta X_t|^2 \le C_T\,\mathbb{E}\int_0^T \Big(|\delta b_s|^2 + |\delta\sigma_s|_F^2 + \|\delta h_s\|^2_{L^2(\nu)}\Big)\mathrm{d}s < \infty$$

by (3.2), so $\delta X \in \mathcal{S}^2(0,T;\mathbb{R}^n)$.

For the backward equation,

$$\delta Y_t = \delta g_T + \int_t^T \delta f_s\,\mathrm{d}s - \int_t^T \delta Z_s\,\mathrm{d}W_s - \int_t^T\int_E \delta U_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e),$$

where $\delta g_T := g(X_T^1, \mathrm{Law}(X_T^1)) - g(X_T^2, \mathrm{Law}(X_T^2))$. As the terminal conditions presuppose $\mathrm{Law}(X_T^k) \in \mathcal{P}_2(\mathbb{R}^n)$, we have $X_T^k \in L^2$, and Assumption 2.3 with (2.9) gives $\mathbb{E}|\delta g_T|^2 \le C\,\mathbb{E}|\delta X_T|^2 < \infty$. Furthermore, (3.2) gives $\mathbb{E}\int_0^T |\delta f_s|^2\,\mathrm{d}s < \infty$. By Lemma 4.1,

$$\mathbb{E}\sup_{t\le T}|\delta Y_t|^2 \le C_{\mathrm{BP}}\Big(\mathbb{E}|\delta g_T|^2 + \mathbb{E}\int_0^T |\delta f_s|^2\,\mathrm{d}s\Big) < \infty, \tag{3.3}$$

therefore $\delta Y \in \mathcal{S}^2(0,T;\mathbb{R}^m)$. By (3.2) and (3.3) we have $\delta\Theta \in \mathbb{S}$.

Now apply Lemma 4.3 of Section 4 to the two solutions at $\lambda = 1$ with $\underline{\lambda} = 1$, zero forcings and the same initial datum: the right-hand side of (4.14) vanishes, so $\|\delta\Theta\|^2_{\mathbb{S}} \le 0$, whence $\Theta^1 = \Theta^2$ and the uniqueness follows. □

Remark 3.4 (Uniqueness without origin integrability). Remark 3.2 would place each tuple $\Theta \in \mathbb{H}^2$ satisfying (1.1) individually in $\mathbb{S}$ if Assumption 2.2 were imposed. But the uniqueness only requires Assumptions 2.3–2.5, as we only need to show that the difference $\Theta^1 - \Theta^2$ belongs to $\mathbb{S}$, using only the Lipschitz control of coefficient differences. However, the existence requires Assumption 2.2.

Theorem 3.5 (Global well-posedness). *For every horizon $T > 0$, under Assumptions 2.2–2.5 the system (1.1) admits a solution $\Theta = (X, Y, Z, U) \in \mathbb{S}$, unique by Theorem 3.3 and* *depending continuously on the coefficients and data in the sense of the estimate* (3.1) *of Theorem 3.1.*

The proof, by monotone continuation from a small-coupling base case, is in Section 4.6.

Corollary 3.6 (Robustness of well-posedness)**.** *Let $m = n$, and let the baseline system $(b_0, \sigma_0, h_0, f_0, g)$, with the initial datum ξ_0, satisfy Assumptions 2.2–2.5 with margin $c_{\text{diss}} > 0$ in the unweighted form* (2.13)*. Let the perturbations p_b, p_σ, p_h, p_f satisfy the corresponding origin-integrability and diagonal Lipschitz conditions of Assumptions 2.2–2.3, and, for some $c_{\text{pert}} \geq 0$, for a.e. $t \in [0, T]$ and all $(\Theta, \mu), (\Theta', \mu') \in \mathcal{D}_t$,*

$$\mathbb{E}\Big[\langle \delta p_b,\, G^\top \delta Y\rangle + \langle \delta p_\sigma,\, G^\top \delta Z\rangle_F + \langle -\delta p_f,\, G\delta X\rangle + \int_E \langle \delta p_h(e),\, G^\top \delta U(e)\rangle \nu(\mathrm{d}e)\Big] \\ \leq c_{\text{pert}}\, \mathbb{E}\Big[|\delta X|^2 + |\delta Y|^2 + |\delta Z|_F^2 + \|\delta U\|_{L^2(\nu)}^2\Big] \quad (3.4)$$

If $c_{\text{pert}} < c_{\text{diss}}$, the perturbed system

$$(b, \sigma, h, f, g) := (b_0 + p_b,\ \sigma_0 + p_\sigma,\ h_0 + p_h,\ f_0 + p_f,\ g)$$

satisfies, with the same initial datum ξ_0, Assumptions 2.2–2.5 with the same $G, \alpha, \beta_1, \beta_2$ and margin $c_{\text{diss}} - c_{\text{pert}}$. By Theorems 3.1, 3.3 and 3.5 it is well-posed.

If, in addition, the perturbations are independent of θ and W_2-Lipschitz,

$$|\delta p_b| + |\delta p_\sigma|_F + \|\delta p_h\|_{L^2(\nu)} + |\delta p_f| \ \leq\ L_p\, W_2(\mu, \mu') \quad (3.5)$$

for a.e. $t \in [0, T]$ and all $(\Theta, \mu), (\Theta', \mu') \in \mathcal{D}_t$, then (3.4) *holds with $c_{\text{pert}} = \|G\| L_p$.*

Proof. Origin values add in L^2 and Lipschitz constants add, so the perturbed system satisfies Assumptions 2.2–2.3; Assumption 2.4 and the terminal inequality of Assumption 2.5 concern only G and g, which are unchanged. For the interior inequality, $\delta\psi = \delta\psi_0 + \delta p_\psi$, $\psi \in \{b, \sigma, h, f\}$, along any pair of diagonal inputs. The pairing splits into a baseline bracket and a perturbation bracket, bounded by (2.13) and (3.4) respectively, off the union of their two Lebesgue-null exceptional time sets:

$$\begin{aligned}
&\mathbb{E}\Big[\langle \delta b,\, G^\top \delta Y\rangle + \langle \delta\sigma,\, G^\top \delta Z\rangle_F + \langle -\delta f,\, G\delta X\rangle + \int_E \langle \delta h(e),\, G^\top \delta U(e)\rangle \nu(\mathrm{d}e)\Big] \\
&\quad = \mathbb{E}\Big[\langle \delta b_0,\, G^\top \delta Y\rangle + \langle \delta\sigma_0,\, G^\top \delta Z\rangle_F + \langle -\delta f_0,\, G\delta X\rangle + \int_E \langle \delta h_0(e),\, G^\top \delta U(e)\rangle \nu(\mathrm{d}e)\Big] \\
&\qquad + \mathbb{E}\Big[\langle \delta p_b,\, G^\top \delta Y\rangle + \langle \delta p_\sigma,\, G^\top \delta Z\rangle_F + \langle -\delta p_f,\, G\delta X\rangle + \int_E \langle \delta p_h(e),\, G^\top \delta U(e)\rangle \nu(\mathrm{d}e)\Big] \\
&\quad \leq -\beta_1 \mathbb{E}\Big[|G^\top \delta Y|^2 + |G^\top \delta Z|_F^2 + \|G^\top \delta U\|_{L^2(\nu)}^2\Big] - \beta_2 \mathbb{E}|G\delta X|^2 \\
&\qquad - (c_{\text{diss}} - c_{\text{pert}})\, \mathbb{E}\Big[|\delta Y|^2 + |\delta Z|_F^2 + \|\delta U\|_{L^2(\nu)}^2 + |\delta X|^2\Big],
\end{aligned}$$

which is (2.13) with the baseline constants and margin $c_{\text{diss}} - c_{\text{pert}}$.

We now prove that (3.5) implies (3.4) with $c_{\mathrm{pert}} = \|G\| L_p$. Write

$$D^2 := \mathbb{E}\Big[|\delta X|^2 + |\delta Y|^2 + |\delta Z|_F^2 + \|\delta U\|^2_{L^2(\nu)}\Big].$$

The differences δp_ψ, $\psi \in \{b, \sigma, h, f\}$, are deterministic, so they factor out of the expectations. Cauchy–Schwarz in each pairing, then (3.5) and (2.8), give

$$\begin{aligned}
&\mathbb{E}\Big[\langle \delta p_b,\, G^\top \delta Y\rangle + \langle \delta p_\sigma,\, G^\top \delta Z\rangle_F + \langle -\delta p_f,\, G\delta X\rangle + \int_E \langle \delta p_h(e),\, G^\top \delta U(e)\rangle \nu(\mathrm{d}e)\Big] \\
&\quad \le \|G\|\, D\, \Big(|\delta p_b| + |\delta p_\sigma|_F + \|\delta p_h\|_{L^2(\nu)} + |\delta p_f|\Big) \\
&\quad \le \|G\|\, D\, L_p\, W_2(\mu, \mu') \\
&\quad \le \|G\| L_p\, D^2,
\end{aligned}$$

which is (3.4) with $c_{\mathrm{pert}} = \|G\| L_p$. □

The margin hypothesis is demanding in the jump channel: by the reduction of Remark 2.10, a margin in (2.13) forces every channel of the baseline to be strictly dissipative. An additive $h_0 = h_0(t, x, e)$ can therefore never serve as a baseline, and jump feedback entering through finitely many functionals of u supplies no margin either; a margin needs feedback dissipative on the whole of $L^2(\nu)$, as in Examples 5.1(i) and 5.5.

Remark 3.7 (General rank). The restriction to $m = n$ reflects only the unweighted bookkeeping: the same splitting of the pairing into baseline and perturbation brackets, run in the weighted quantities of (2.12), gives the identical conclusion for arbitrary m, n when the margin and (3.4) are both recorded in those quantities.

Proposition 3.8 (Strictness of expected diagonal monotonicity)**.** *Let $n = m = 1$, $d \ge 1$ and $G = 1$, and fix $\beta_1 > 0$. For every $0 < \varepsilon < \beta_1$, the drift*

$$b(t, \theta, \mu) := -\beta_1 y + \varepsilon\, g_0\Big(y - m_Y(\mu)\Big), \qquad g_0(r) := \max\{-1, \min\{1, r\}\}, \tag{3.6}$$

with $m_Y(\mu) := \int_{\mathcal{H}} y\, \mu(\mathrm{d}\theta)$ the Y-marginal mean, is free-measure W_2-Lipschitz with constant $\beta_1 + \varepsilon$ and satisfies, along all $(\Theta, \mu), (\Theta', \mu') \in \mathcal{D}_t$,

$$\mathbb{E}\langle \delta b,\, \delta Y\rangle \;\le\; -(\beta_1 - \varepsilon)\, \mathbb{E}|\delta Y|^2; \tag{3.7}$$

yet for every $\beta \ge 0$ the pointwise bound $\langle \delta b,\, \delta y\rangle \le -\beta\, |\delta y|^2$ fails at a pair of free inputs with $\theta \in \operatorname{supp} \mu$ and $\theta' \in \operatorname{supp} \mu'$.

Proof. Free-measure Lipschitz. Since g_0 is 1-Lipschitz and by (2.10),

$$\begin{aligned}
|\delta b| \;&\le\; \beta_1 |\delta y| + \varepsilon \Big|\Big(y - m_Y(\mu)\Big) - \Big(y' - m_Y(\mu')\Big)\Big| \\
&\le\; \beta_1 |\delta y| + \varepsilon \Big(|\delta y| + |m_Y(\mu) - m_Y(\mu')|\Big) \\
&\le\; (\beta_1 + \varepsilon) \Big(|\delta y| + W_2(\mu, \mu')\Big).
\end{aligned}$$

Expected diagonal bound. Along diagonal inputs, $m_Y(\mu) = \mathbb{E}Y$ and $m_Y(\mu') = \mathbb{E}Y'$. Since g_0 is 1-Lipschitz and its arguments differ by $(Y - \mathbb{E}Y) - (Y' - \mathbb{E}Y') = \delta Y - \mathbb{E}\,\delta Y$, Cauchy–Schwarz and $\mathbb{E}|\delta Y - \mathbb{E}\,\delta Y|^2 = \mathbb{E}|\delta Y|^2 - |\mathbb{E}\,\delta Y|^2 \le \mathbb{E}|\delta Y|^2$ give

$$\begin{aligned}
\mathbb{E}\langle \delta b,\, \delta Y\rangle &= -\beta_1 \mathbb{E}|\delta Y|^2 + \varepsilon\, \mathbb{E}\Big[\Big(g_0(Y - \mathbb{E}Y) - g_0(Y' - \mathbb{E}Y')\Big)\, \delta Y\Big] \\
&\le -\beta_1 \mathbb{E}|\delta Y|^2 + \varepsilon \Big(\mathbb{E}|\delta Y - \mathbb{E}\,\delta Y|^2\Big)^{1/2} \Big(\mathbb{E}|\delta Y|^2\Big)^{1/2} \\
&\le -(\beta_1 - \varepsilon)\, \mathbb{E}|\delta Y|^2,
\end{aligned}$$

which is (3.7).

Pointwise failure on the supports. Take $\theta = (0, y, 0, 0)$ and $\theta' = (0, y - t, 0, 0)$ with $t > 0$, and the two-point laws

$$\mu := \tfrac{1}{2}\delta_{(0,y,0,0)} + \tfrac{1}{2}\delta_{(0,y-2,0,0)}, \qquad \mu' := \tfrac{1}{2}\delta_{(0,y-t,0,0)} + \tfrac{1}{2}\delta_{(0,y-t+2,0,0)},$$

so that $\theta \in \operatorname{supp}\mu$, $\theta' \in \operatorname{supp}\mu'$, and $m_Y(\mu) = y - 1$, $m_Y(\mu') = y - t + 1$. The two arguments of g_0 are then ± 1, and

$$\begin{gathered}
y - m_Y(\mu) = 1, \qquad (y - t) - m_Y(\mu') = -1, \qquad \delta y = t, \\
\delta b = -\beta_1 t + \varepsilon\Big(g_0(1) - g_0(-1)\Big) = -\beta_1 t + 2\varepsilon, \\
\langle \delta b,\, \delta y\rangle = -\beta_1 t^2 + 2\varepsilon t > 0 \qquad \text{for } 0 < t < 2\varepsilon/\beta_1,
\end{gathered}$$

incompatible with every nonpositive bound $-\beta t^2$, $\beta \ge 0$. $\square$

4 Proof of well-posedness

The route is: backward L^2 estimate (Lemma 4.1) $\to$ small-coupling base case (Lemma 4.2) $\to$ uniform stability on $[\lambda_*, 1]$ (Lemma 4.3), which at $\lambda = 1$ gives Theorems 3.1 and 3.3 $\to$ fixed-increment continuation (Lemma 4.4) $\to$ Theorem 3.5, the continuation being along the coupling strength, in the tradition of Peng and Wu [18]. The central a priori estimate rests on a mixed G-energy identity for $\langle G\,\delta X_t,\, \delta Y_t\rangle$, whose compensated-Poisson cross-variation produces a coercive $L^2(\nu)$ jump term controlled by the G-monotonicity assumption. The continuation is in the coupling strength, not the horizon: shortening the horizon does not make the frozen-coefficient map contractive, since the (z, u)-dependence of σ, h and the full-tuple law coupling enter the contraction constant with no T-factor.

Inside $\mathrm{d}t$-integrals we do not distinguish X_t, Y_t from X_{t-}, Y_{t-}: càdlàg paths have only countably many jumps on compact intervals, so $\Theta_t^- = \Theta_t$ $\mathbb{P}$-a.s. and $\mu_t^- = \mu_t$ for a.e. t (Lemma A.1).

Throughout this section, C denotes a positive constant, not necessarily the same at each occurrence, depending only on $T, L, \|G\|, c_G, \alpha, \beta_1, \beta_2$ and the BDG constant c_{BDG}, unless a different dependence is stated; recurring constants are subscripted (C_0, C_{BP}, C_B, …) and fixed where they are introduced.

Itô's formula for Lévy-type stochastic integrals, and the product form obtained by applying it to the bilinear map $(x,y)\mapsto\langle Gx,\,y\rangle$, are those of [1, §4.4], stated for the Brownian–Poisson setting fixed above. The maximal estimates use the Burkholder–Davis–Gundy inequality at $p=1$ in quadratic-variation form ([9], chapter on H^1 and BMO); valid for càdlàg local martingales, it covers the compensated-Poisson martingales below without a separate jump statement. By localisation and dominated convergence, a càdlàg local martingale whose running supremum is integrable has zero-mean increments (the *supremum criterion* below).

For $\lambda\in[0,1]$ we work throughout with the λ-*scaled, forced* system: for a square-integrable forcing $q=(q^b,q^\sigma,q^h,q^f,q^g_T)$ with

$$
\begin{aligned}
&q^b\in L^2(\Omega\times[0,T];\mathbb{R}^n), && q^\sigma\in L^2(\Omega\times[0,T];\mathbb{R}^{n\times d}),\\
&q^h\in L^2(\Omega\times[0,T]\times E;\mathbb{R}^n), && q^f\in L^2(\Omega\times[0,T];\mathbb{R}^m),\\
&q^g_T\in L^2(\Omega,\mathcal{F}_T;\mathbb{R}^m),
\end{aligned}
$$

where q^b,q^σ,q^f are progressively measurable and q^h is predictable ($\mathcal{P}\otimes\mathcal{E}$-measurable as in (2.4), square-integrable for $\mathrm{d}\mathbb{P}\otimes\mathrm{d}t\otimes\nu$),

$$
\begin{cases}
X_t=\xi_0+\lambda\displaystyle\int_0^t b(s,\Theta_s,\mu_s)\,\mathrm{d}s+\lambda\int_0^t\sigma(s,\Theta_s,\mu_s)\,\mathrm{d}W_s\\
\qquad\qquad+\lambda\displaystyle\int_0^t\int_E h(s,\Theta_s^-,e,\mu_s^-)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e)\\
\qquad\qquad+\displaystyle\int_0^t q^b_s\,\mathrm{d}s+\int_0^t q^\sigma_s\,\mathrm{d}W_s+\int_0^t\int_E q^h_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e),\\[2ex]
Y_t=\lambda\,g(X_T,\mathrm{Law}(X_T))+\lambda\displaystyle\int_t^T f(s,\Theta_s,\mu_s)\,\mathrm{d}s+q^g_T+\int_t^T q^f_s\,\mathrm{d}s\\
\qquad\qquad-\displaystyle\int_t^T Z_s\,\mathrm{d}W_s-\int_t^T\int_E U_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e).
\end{cases}
\tag{4.1}
$$

Call the system (4.1) *well posed at* λ — written $\mathsf{WP}(\lambda)$ — if it has a unique solution in $\mathbb{S}$ for *every* such forcing q. The unforced target is $\mathsf{WP}(1)$ with $q\equiv 0$, so it suffices to prove that $\mathsf{WP}(1)$ holds.

By Assumptions 2.2–2.3, the coefficient processes (2.5) along any $\Theta\in\mathbb{S}$ are square-integrable: for $\psi\in\{b,\sigma,h,f\}$, in the norms of Assumption 2.3,

$$
\mathbb{E}\int_0^T|\psi(s,\Theta_s,\mu_s)|^2\,\mathrm{d}s\le C\,\mathbb{E}\int_0^T\Big(|X_s|^2+|Y_s|^2+|Z_s|_F^2+\|U_s\|^2_{L^2(\nu)}+|\psi(s,0,\delta_0)|^2\Big)\mathrm{d}s,
\tag{4.2}
$$

by that Lipschitz bound together with the coupling estimate (2.8) at $\mu'=\delta_0$. For the terminal function the analogous bound

$$\mathbb{E}|g(X_T, \mathrm{Law}(X_T))|^2 \leq C\Big(\mathbb{E}|X_T|^2 + |g(0, \delta_0)|^2\Big) \tag{4.3}$$

follows in the same way from (2.6) and (2.9).

4.1 A Brownian–Poisson L^2 estimate

Lemma 4.1 (Brownian–Poisson L^2 estimate)**.** *Let δY be a càdlàg adapted process and let $\delta Z \in \mathcal{H}^2_W$, $\delta U \in \mathcal{H}^2_N$ satisfy*

$$\delta Y_t = \delta\xi + \int_t^T \phi_s \,\mathrm{d}s - \int_t^T \delta Z_s \,\mathrm{d}W_s - \int_t^T \int_E \delta U_s(e)\, \widetilde{N}(\mathrm{d}s, \mathrm{d}e) \tag{4.4}$$

with $\delta\xi \in L^2(\mathcal{F}_T)$ and $\phi \in L^2(\Omega \times [0,T])$. Then $\delta Y \in \mathcal{S}^2(0,T;\mathbb{R}^m)$, and there is a constant C_{BP}, depending only on T and c_{BDG}, with

$$\mathbb{E}\sup_{t\leq T} |\delta Y_t|^2 + \mathbb{E}\int_0^T \Big(|\delta Z_s|_F^2 + \|\delta U_s\|^2_{L^2(\nu)}\Big)\mathrm{d}s \leq C_{\mathrm{BP}}\Big(\mathbb{E}|\delta\xi|^2 + \mathbb{E}\int_0^T |\phi_s|^2 \mathrm{d}s\Big).$$

Proof. Membership. By (4.4) with $\int_t^T = \int_0^T - \int_0^t$,

$$\sup_{t\leq T} |\delta Y_t| \leq |\delta\xi| + \int_0^T |\phi_s| \,\mathrm{d}s + 2\sup_{t\leq T}\Big|\int_0^t \delta Z_s \,\mathrm{d}W_s\Big| + 2\sup_{t\leq T}\Big|\int_0^t\int_E \delta U_s(e)\, \widetilde{N}(\mathrm{d}s, \mathrm{d}e)\Big|.$$

Both stochastic integrals are true L^2-martingales by the Itô and compensated-Poisson isometries, as $\delta Z \in \mathcal{H}^2_W$ and $\delta U \in \mathcal{H}^2_N$. Squaring, Cauchy–Schwarz on the drift term, and Doob's inequality then give $\mathbb{E}\sup_{t\leq T} |\delta Y_t|^2 < \infty$, that is, $\delta Y \in \mathcal{S}^2(0,T;\mathbb{R}^m)$.

The $\mathrm{d}s$- and $\mathrm{d}W$-integrals of δY are continuous, so δY jumps only through its compensated Poisson integral: its jumps $\Delta\, \delta Y_s := \delta Y_s - \delta Y_{s-}$ satisfy $\Delta\, \delta Y_s = \delta U_s(e)$ at each atom (s, e) of N; hence the jump variation splits through $N = \widetilde{N} + \mathrm{d}s\,\nu$ as

$$\begin{aligned}\sum_{t<s\leq T} |\Delta\, \delta Y_s|^2 &= \int_t^T\int_E |\delta U_s(e)|^2\, N(\mathrm{d}s, \mathrm{d}e) \\ &= \int_t^T\int_E |\delta U_s(e)|^2\, \widetilde{N}(\mathrm{d}s, \mathrm{d}e) + \int_t^T \|\delta U_s\|^2_{L^2(\nu)}\,\mathrm{d}s.\end{aligned}$$

Itô's formula for $e^t|\delta Y_t|^2$ then gives the pathwise identity

$$
\begin{aligned}
e^t|\delta Y_t|^2 &+ \int_t^T e^s\Big(|\delta Y_s|^2 + |\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s \\
&\quad = e^T|\delta\xi|^2 + 2\int_t^T e^s\langle \delta Y_s,\, \phi_s\rangle \mathrm{d}s \\
&\qquad - 2\int_t^T e^s\langle \delta Y_{s-},\, \delta Z_s\,\mathrm{d}W_s\rangle \\
&\qquad - 2\int_t^T\int_E e^s\langle \delta Y_{s-},\, \delta U_s(e)\rangle\, \widetilde{N}(\mathrm{d}s,\mathrm{d}e) \\
&\qquad - \int_t^T\int_E e^s|\delta U_s(e)|^2\, \widetilde{N}(\mathrm{d}s,\mathrm{d}e).
\end{aligned}
\tag{4.5}
$$

Zero-mean terms. The stochastic-integral terms in (4.5) decompose into three local martingales

$$
\begin{aligned}
M_t^W &:= \int_0^t e^s\langle \delta Y_{s-},\, \delta Z_s\,\mathrm{d}W_s\rangle, \\
M_t^1 &:= \int_0^t\int_E e^s\langle \delta Y_{s-},\, \delta U_s(e)\rangle\, \widetilde{N}(\mathrm{d}s,\mathrm{d}e), \\
M_t^2 &:= \int_0^t\int_E e^s|\delta U_s(e)|^2\, \widetilde{N}(\mathrm{d}s,\mathrm{d}e),
\end{aligned}
$$

each of which we now show satisfies $\mathbb{E}[M_T - M_t] = 0$. The three do not share one argument: $\delta U \in \mathcal{H}_N^2$ gives the quadratic integrand of M^2 no square moments, so the L^2 theory behind the bounds below does not reach it — nor, with $\nu(E) = \infty$ admitted, does localisation along jump times, which then fail to exhaust $[0,T]$. M^2 is instead handled through the compensation formula, which asks only for integrability. For M^W, BDG and Cauchy–Schwarz give

$$
\mathbb{E}\sup_{t\le T}|M_t^W| \le c_{\mathrm{BDG}}\, e^T\Big(\mathbb{E}\sup_{t\le T}|\delta Y_t|^2\Big)^{1/2}\Big(\mathbb{E}\int_0^T|\delta Z_s|_F^2\,\mathrm{d}s\Big)^{1/2} < \infty,
$$

and for M^1 BDG, Cauchy–Schwarz and (2.1) give

$$
\begin{aligned}
\mathbb{E}\sup_{t\le T}|M_t^1| &\le c_{\mathrm{BDG}}\,\mathbb{E}\Big[\Big(\int_0^T\int_E e^{2s}|\delta Y_{s-}|^2|\delta U_s(e)|^2\,N(\mathrm{d}s,\mathrm{d}e)\Big)^{1/2}\Big] \\
&\le c_{\mathrm{BDG}}\, e^T\Big(\mathbb{E}\sup_{t\le T}|\delta Y_t|^2\Big)^{1/2}\Big(\mathbb{E}\int_0^T\int_E|\delta U_s(e)|^2\,N(\mathrm{d}s,\mathrm{d}e)\Big)^{1/2} \\
&= c_{\mathrm{BDG}}\, e^T\Big(\mathbb{E}\sup_{t\le T}|\delta Y_t|^2\Big)^{1/2}\Big(\mathbb{E}\int_0^T\int_E|\delta U_s(e)|^2\,\nu(\mathrm{d}e)\,\mathrm{d}s\Big)^{1/2} \\
&= c_{\mathrm{BDG}}\, e^T\Big(\mathbb{E}\sup_{t\le T}|\delta Y_t|^2\Big)^{1/2}\Big(\mathbb{E}\int_0^T\|\delta U_s\|_{L^2(\nu)}^2\,\mathrm{d}s\Big)^{1/2} < \infty.
\end{aligned}
$$

These bounds put the running suprema of M^W and M^1 in L^1, so the supremum criterion gives $\mathbb{E}[M_T - M_t] = 0$ for $M \in \{M^W, M^1\}$. For M^2 the integrand $e^s|\delta U_s(e)|^2$ is predictable and nonnegative and, because $\delta U \in \mathcal{H}^2_N$, lies in $L^1(\mathrm{d}\mathbb{P}\otimes \mathrm{d}s\otimes\nu)$:

$$\mathbb{E}\int_0^T\int_E e^s|\delta U_s(e)|^2\,\nu(\mathrm{d}e)\,\mathrm{d}s \le e^T\,\mathbb{E}\int_0^T\|\delta U_s\|^2_{L^2(\nu)}\,\mathrm{d}s<\infty.$$

Both the Poisson integral and its compensator are therefore integrable, so the increment splits under the expectation:

$$\begin{aligned}\mathbb{E}[M_T^2-M_t^2] &= \mathbb{E}\int_t^T\int_E e^s|\delta U_s(e)|^2\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e)\\ &= \mathbb{E}\int_t^T\int_E e^s|\delta U_s(e)|^2\,N(\mathrm{d}s,\mathrm{d}e) - \mathbb{E}\int_t^T\int_E e^s|\delta U_s(e)|^2\,\nu(\mathrm{d}e)\,\mathrm{d}s\\ &= 0.\end{aligned}$$

Taking expectations in (4.5) therefore removes the three martingale terms, and bounding the drift term by Young's inequality, $2\langle\delta Y_s,\,\phi_s\rangle\le|\delta Y_s|^2+|\phi_s|^2$, cancels the $|\delta Y_s|^2$ integrals on the two sides:

$$\mathbb{E}e^t|\delta Y_t|^2+\mathbb{E}\int_t^T e^s\Big(|\delta Z_s|_F^2+\|\delta U_s\|^2_{L^2(\nu)}\Big)\mathrm{d}s\le \mathbb{E}e^T|\delta\xi|^2+\mathbb{E}\int_t^T e^s|\phi_s|^2\mathrm{d}s.$$

Bounding $1\le e^s\le e^T$ on both sides leaves, for every $t\le T$,

$$\mathbb{E}|\delta Y_t|^2+\mathbb{E}\int_t^T\Big(|\delta Z_s|_F^2+\|\delta U_s\|^2_{L^2(\nu)}\Big)\mathrm{d}s\le R,\qquad R:=e^T\Big(\mathbb{E}|\delta\xi|^2+\mathbb{E}\int_0^T|\phi_s|^2\mathrm{d}s\Big).$$

Both terms on the left are nonnegative and R does not depend on t, so each is bounded by R on its own, the first uniformly in t and the second at $t=0$; adding them,

$$\begin{aligned}&\sup_{t\le T}\mathbb{E}|\delta Y_t|^2+\mathbb{E}\int_0^T\Big(|\delta Z_s|_F^2+\|\delta U_s\|^2_{L^2(\nu)}\Big)\mathrm{d}s\\ &\qquad\le 2R = C_0\Big(\mathbb{E}|\delta\xi|^2+\mathbb{E}\int_0^T|\phi_s|^2\mathrm{d}s\Big),\qquad C_0:=2e^T.\end{aligned}\tag{4.6}$$

It remains to upgrade $\sup_{t\le T}\mathbb{E}|\delta Y_t|^2$ to $\mathbb{E}\sup_{t\le T}|\delta Y_t|^2$. Taking the difference of (4.4) at times t and 0 gives the forward form

$$\delta Y_t=\delta Y_0-\int_0^t\phi_s\mathrm{d}s+\int_0^t\delta Z_s\mathrm{d}W_s+\int_0^t\int_E\delta U_s(e)\widetilde{N}(\mathrm{d}s,\mathrm{d}e).$$

Squaring, taking the supremum over t, Cauchy–Schwarz on the ds-integral, and BDG on the two martingales give

$$\begin{aligned}\mathbb{E}\sup_{t\le T}|\delta Y_t|^2 &\le 4\,\mathbb{E}|\delta Y_0|^2+4\,\mathbb{E}\Big(\int_0^T|\phi_s|\mathrm{d}s\Big)^2\\ &\qquad+4\,\mathbb{E}\sup_{t\le T}\Big|\int_0^t\delta Z_s\mathrm{d}W_s\Big|^2+4\,\mathbb{E}\sup_{t\le T}\Big|\int_0^t\int_E\delta U_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e)\Big|^2\\ &\le 4\,\mathbb{E}|\delta Y_0|^2+4T\,\mathbb{E}\int_0^T|\phi_s|^2\mathrm{d}s+4c_{\mathrm{BDG}}\,\mathbb{E}\int_0^T\Big(|\delta Z_s|_F^2+\|\delta U_s\|^2_{L^2(\nu)}\Big)\mathrm{d}s.\end{aligned}$$

Using $\mathbb{E}|\delta Y_0|^2 \le \sup_{t\le T}\mathbb{E}|\delta Y_t|^2$ and (4.6),

$$\begin{aligned}
\mathbb{E}\sup_{t\le T}|\delta Y_t|^2 &+ \mathbb{E}\int_0^T \Big(|\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s \\
&\le 4\sup_{t\le T}\mathbb{E}|\delta Y_t|^2 + (4c_{\mathrm{BDG}}+1)\,\mathbb{E}\int_0^T \Big(|\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s \\
&\qquad + 4T\,\mathbb{E}\int_0^T |\phi_s|^2\mathrm{d}s \\
&\le \Big(4C_0 + (4c_{\mathrm{BDG}}+1)C_0 + 4T\Big)\Big(\mathbb{E}|\delta\xi|^2 + \mathbb{E}\int_0^T|\phi_s|^2\mathrm{d}s\Big) \\
&= C_{\mathrm{BP}}\Big(\mathbb{E}|\delta\xi|^2 + \mathbb{E}\int_0^T|\phi_s|^2\mathrm{d}s\Big), \qquad C_{\mathrm{BP}} := C_0(5+4c_{\mathrm{BDG}}) + 4T.
\end{aligned}$$

This depends only on T and c_{BDG}. $\square$

4.2 Small-coupling solvability

Lemma 4.2 (Small-coupling base case). *Under Assumptions 2.2–2.3, there is $\lambda_* \in (0,1]$, depending only on T, L and c_{BDG}, such that $\mathsf{WP}(\lambda)$ holds for every $\lambda\in[0,\lambda_*]$.*

Proof. Fix $\lambda\in[0,1]$ and a forcing q. For a trial tuple $\Theta=(X,Y,Z,U)\in\mathbb{S}$ write

$$\mu_t^{\Theta} := \mathrm{Law}(\Theta_t), \qquad \mu_t^{\Theta,-} := \mathrm{Law}(\Theta_t^-),$$

and define $\Phi_\lambda(\Theta) := \bar\Theta = (\bar X,\bar Y,\bar Z,\bar U)$ by the forward and backward constructions below.

Forward. Let $\bar X$ be given by the frozen forward dynamics

$$\begin{aligned}
\bar X_t = \xi_0 + \lambda\int_0^t b(s,\Theta_s,\mu_s^\Theta)\,\mathrm{d}s + \lambda\int_0^t \sigma(s,\Theta_s,\mu_s^\Theta)\,\mathrm{d}W_s \\
+ \lambda\int_0^t\!\!\int_E h(s,\Theta_s^-,e,\mu_s^{\Theta,-})\,\widetilde N(\mathrm{d}s,\mathrm{d}e) \\
+ \int_0^t q_s^b\,\mathrm{d}s + \int_0^t q_s^\sigma\,\mathrm{d}W_s + \int_0^t\!\!\int_E q_s^h(e)\,\widetilde N(\mathrm{d}s,\mathrm{d}e).
\end{aligned}\tag{4.7}$$

In (4.7) the coefficient h is evaluated at the left limits $(\Theta_s^-,\mu_s^{\Theta,-})$; by Lemma A.1 these agree with (Θ_s,μ_s^Θ) outside a $\mathrm{d}t\otimes\mathrm{d}\mathbb{P}$-null set, so either form may be used inside the $\mathrm{d}t\otimes\mathrm{d}\mathbb{P}$-integrals below. Since $\Theta\in\mathbb{S}$, (4.2) gives

$$\mathbb{E}\int_0^T|\psi(s,\Theta_s,\mu_s^\Theta)|^2\,\mathrm{d}s<\infty \qquad \text{for } \psi\in\{b,\sigma,h\},$$

and q^b,q^σ,q^h are square-integrable by hypothesis. The coefficients are jointly Borel, so these integrands are progressively measurable and the jump integrand is predictable. Cauchy–Schwarz on the drift and Doob/BDG on the two martingale integrals then give $\mathbb{E}\sup_{t\le T}|\bar X_t|^2<\infty$, so (4.7) defines a unique $\bar X\in\mathcal{S}^2(0,T;\mathbb{R}^n)$.

Backward. The random variable

$$\Xi := \lambda\, g(\bar X_T, \mathrm{Law}(\bar X_T)) + q_T^g + \int_0^T (\lambda f(s, \Theta_s, \mu_s^\Theta) + q_s^f)\,\mathrm{d}s$$

lies in $L^2(\Omega, \mathcal{F}_T; \mathbb{R}^m)$ by (4.3), (4.2) applied to f, and the square-integrability of q_T^g and q^f. Set $M_t := \mathbb{E}[\Xi \mid \mathcal{F}_t]$. By the Brownian–Poisson predictable representation property there are unique $\bar Z \in \mathcal{H}_W^2$, $\bar U \in \mathcal{H}_N^2$ with

$$M_t = M_0 + \int_0^t \bar Z_s\,\mathrm{d}W_s + \int_0^t \int_E \bar U_s(e)\,\widetilde N(\mathrm{d}s, \mathrm{d}e).$$

Put

$$\bar Y_t := M_t - \int_0^t (\lambda f(s, \Theta_s, \mu_s^\Theta) + q_s^f)\,\mathrm{d}s.$$

Doob's inequality and $\Xi \in L^2$ give $M \in \mathcal{S}^2(0, T; \mathbb{R}^m)$. The drift is square-integrable by (4.2) and $q^f \in L^2$, so Cauchy–Schwarz gives $\bar Y \in \mathcal{S}^2(0, T; \mathbb{R}^m)$. By construction $(\bar Y, \bar Z, \bar U)$ solves the frozen linear BSDE

$$\begin{aligned}\bar Y_t = \lambda g(\bar X_T, \mathrm{Law}(\bar X_T)) + q_T^g + \int_t^T (\lambda f(s, \Theta_s, \mu_s^\Theta) + q_s^f)\,\mathrm{d}s \\ - \int_t^T \bar Z_s\,\mathrm{d}W_s - \int_t^T \int_E \bar U_s(e)\,\widetilde N(\mathrm{d}s, \mathrm{d}e),\end{aligned} \tag{4.8}$$

so $\bar\Theta \in \mathbb{S}$ and $\Phi_\lambda : \mathbb{S} \to \mathbb{S}$ is well defined.

Contraction. Take $\Theta^1, \Theta^2 \in \mathbb{S}$, write $\bar\Theta^k = \Phi_\lambda(\Theta^k)$, and set

$$\delta\psi_s := \psi(s, \Theta_s^1, \mu_s^{\Theta^1}) - \psi(s, \Theta_s^2, \mu_s^{\Theta^2}) \quad (\psi \in \{b, \sigma, h, f\}),$$
$$\delta X := X^1 - X^2, \qquad \delta \bar X := \bar X^1 - \bar X^2,$$

and likewise for the remaining components of $\Theta^1 - \Theta^2$ and of $\bar\Theta^1 - \bar\Theta^2$; the common initial datum ξ_0 and the fixed forcing q cancel in every difference.

Step 1: the coefficient differences. Write $S_s := |\delta b_s| + |\delta\sigma_s|_F + \|\delta h_s\|_{L^2(\nu)} + |\delta f_s|$. Assumption 2.3, applied to the diagonal inputs $(\Theta_s^1, \mu_s^{\Theta^1})$ and $(\Theta_s^2, \mu_s^{\Theta^2})$, gives

$$S_s \le L\Big(|\delta X_s| + |\delta Y_s| + |\delta Z_s|_F + \|\delta U_s\|_{L^2(\nu)} + W_2\Big(\mu_s^{\Theta^1}, \mu_s^{\Theta^2}\Big)\Big).$$

Squaring, using Cauchy–Schwarz in $\mathbb{R}^5$, taking expectations, and bounding the deterministic Wasserstein term by (2.8) give

$$\mathbb{E}\, S_s^2 \le 10L^2\, \mathbb{E}\Big[|\delta X_s|^2 + |\delta Y_s|^2 + |\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big].$$

Integrating in s and dominating the X- and Y-slots by their suprema,

$$
\begin{aligned}
\mathbb{E}\int_0^T S_s^2\,\mathrm{d}s &\le 10L^2\,\mathbb{E}\int_0^T\Big(|\delta X_s|^2+|\delta Y_s|^2+|\delta Z_s|_F^2+\|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s\\
&\le 10L^2\Big(T\,\mathbb{E}\sup_{t\le T}|\delta X_t|^2+T\,\mathbb{E}\sup_{t\le T}|\delta Y_t|^2\\
&\qquad\qquad+\mathbb{E}\int_0^T\Big(|\delta Z_s|_F^2+\|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s\Big)\\
&\le L_\star^2\,\|\Theta^1-\Theta^2\|_{\mathbb{S}}^2,\qquad L_\star^2:=10L^2\max\{T,1\}.
\end{aligned}
\tag{4.9}
$$

Step 2: the forward difference. Taking the difference of (4.7) at Θ^1 and Θ^2,

$$
\delta\bar{X}_t=\lambda\int_0^t\delta b_s\,\mathrm{d}s+\lambda\int_0^t\delta\sigma_s\,\mathrm{d}W_s+\lambda\int_0^t\int_E\delta h_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e).
$$

Squaring, taking the supremum in t, and taking expectations,

$$
\begin{aligned}
\mathbb{E}\sup_{t\le T}|\delta\bar{X}_t|^2 &\le 3\lambda^2\Big[\mathbb{E}\Big(\int_0^T|\delta b_s|\,\mathrm{d}s\Big)^2+\mathbb{E}\sup_{t\le T}\Big|\int_0^t\delta\sigma_s\,\mathrm{d}W_s\Big|^2\\
&\qquad\qquad+\mathbb{E}\sup_{t\le T}\Big|\int_0^t\int_E\delta h_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e)\Big|^2\Big]\\
&\le 3\lambda^2\Big[T\,\mathbb{E}\int_0^T|\delta b_s|^2\,\mathrm{d}s+c_{\mathrm{BDG}}\,\mathbb{E}\int_0^T|\delta\sigma_s|_F^2\,\mathrm{d}s+c_{\mathrm{BDG}}\,\mathbb{E}\int_0^T\|\delta h_s\|_{L^2(\nu)}^2\,\mathrm{d}s\Big]\\
&\le 3\lambda^2\max\{T,c_{\mathrm{BDG}}\}\,\mathbb{E}\int_0^T\Big(|\delta b_s|^2+|\delta\sigma_s|_F^2+\|\delta h_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s\\
&\le 3\lambda^2\max\{T,c_{\mathrm{BDG}}\}\,\mathbb{E}\int_0^T S_s^2\,\mathrm{d}s\\
&\le C_X\lambda^2\,\|\Theta^1-\Theta^2\|_{\mathbb{S}}^2,\qquad C_X:=3\max\{T,c_{\mathrm{BDG}}\}\,L_\star^2,
\end{aligned}
\tag{4.10}
$$

the jump term by (2.1), the next step since the summands of S_s are non-negative, and the last by (4.9).

Step 3: the terminal difference. Put

$$
\delta\bar{\xi}:=\lambda\Big[g(\bar{X}_T^1,\mathrm{Law}(\bar{X}_T^1))-g(\bar{X}_T^2,\mathrm{Law}(\bar{X}_T^2))\Big].
$$

Then

$$
\begin{aligned}
\mathbb{E}|\delta\bar{\xi}|^2 &\le \lambda^2L^2\,\mathbb{E}\Big(|\delta\bar{X}_T|+W_2(\mathrm{Law}(\bar{X}_T^1),\mathrm{Law}(\bar{X}_T^2))\Big)^2\\
&\le 2\lambda^2L^2\Big(\mathbb{E}|\delta\bar{X}_T|^2+W_2(\mathrm{Law}(\bar{X}_T^1),\mathrm{Law}(\bar{X}_T^2))^2\Big)\\
&\le C_g\lambda^2\,\mathbb{E}|\delta\bar{X}_T|^2,\qquad C_g:=4L^2,
\end{aligned}
\tag{4.11}
$$

the first step by (2.6) and the last by (2.9).

Step 4: the backward difference. Taking the difference of (4.8) at Θ^1 and Θ^2, the terms

q_T^g and q^f cancelling,

$$\delta\bar{Y}_t = \delta\bar{\xi} + \int_t^T \lambda\,\delta f_s\,\mathrm{d}s - \int_t^T \delta\bar{Z}_s\,\mathrm{d}W_s - \int_t^T\int_E \delta\bar{U}_s(e)\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e),$$

which is (4.4) with $\delta\xi = \delta\bar{\xi}$ and $\phi_s = \lambda\,\delta f_s$. $\delta\bar{Y}$ is càdlàg adapted, and the remaining hypotheses hold:

$$\begin{aligned} (\delta\bar{Z},\delta\bar{U}) &\in \mathcal{H}_W^2\times\mathcal{H}_N^2 && \text{since } \bar{\Theta}^1,\bar{\Theta}^2\in\mathbb{S},\\ \delta\bar{\xi} &\in L^2(\mathcal{F}_T) && \text{by (4.3)},\\ \lambda\,\delta f &\in L^2(\Omega\times[0,T]) && \text{by (4.9), as } |\delta f|\le S. \end{aligned}$$

By Lemma 4.1,

$$\begin{aligned} &\mathbb{E}\sup_{t\le T}|\delta\bar{Y}_t|^2 + \mathbb{E}\int_0^T\Big(|\delta\bar{Z}_s|_F^2 + \|\delta\bar{U}_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s\\ &\qquad\le C_{\mathrm{BP}}\Big(\mathbb{E}|\delta\bar{\xi}|^2 + \lambda^2\,\mathbb{E}\int_0^T|\delta f_s|^2\,\mathrm{d}s\Big)\\ &\qquad\le C_{\mathrm{BP}}\Big(C_g\lambda^2\,\mathbb{E}|\delta\bar{X}_T|^2 + L_\star^2\lambda^2\,\|\Theta^1-\Theta^2\|_{\mathbb{S}}^2\Big)\\ &\qquad\le C_{\mathrm{BP}}\Big(C_g\lambda^2\,\mathbb{E}\sup_{t\le T}|\delta\bar{X}_t|^2 + L_\star^2\lambda^2\,\|\Theta^1-\Theta^2\|_{\mathbb{S}}^2\Big)\\ &\qquad\le C_{\mathrm{BP}}\Big(C_gC_X\lambda^4 + L_\star^2\lambda^2\Big)\|\Theta^1-\Theta^2\|_{\mathbb{S}}^2\\ &\qquad\le C_Y\lambda^2\,\|\Theta^1-\Theta^2\|_{\mathbb{S}}^2, \qquad C_Y := C_{\mathrm{BP}}\Big(C_gC_X + L_\star^2\Big), \end{aligned}\tag{4.12}$$

the second inequality by (4.11) and (4.9), the fourth by (4.10), and the last by $\lambda^4\le\lambda^2$ for $\lambda\in[0,1]$.

Step 5: the contraction. Since $\Phi_\lambda(\Theta^1)-\Phi_\lambda(\Theta^2) = (\delta\bar{X},\delta\bar{Y},\delta\bar{Z},\delta\bar{U})$,

$$\begin{aligned} \|\Phi_\lambda(\Theta^1)-\Phi_\lambda(\Theta^2)\|_{\mathbb{S}}^2 &= \mathbb{E}\sup_{t\le T}|\delta\bar{X}_t|^2\\ &\qquad + \mathbb{E}\sup_{t\le T}|\delta\bar{Y}_t|^2 + \mathbb{E}\int_0^T\Big(|\delta\bar{Z}_s|_F^2 + \|\delta\bar{U}_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s\\ &\le (C_X+C_Y)\lambda^2\,\|\Theta^1-\Theta^2\|_{\mathbb{S}}^2\\ &\le C_\Phi\lambda^2\,\|\Theta^1-\Theta^2\|_{\mathbb{S}}^2, \qquad C_\Phi := \max\{1, C_X+C_Y\}, \end{aligned}\tag{4.13}$$

the first inequality by (4.10) and (4.12). Here

$$L_\star = L_\star(T,L),\quad C_X = C_X(T,L,c_{\mathrm{BDG}}),\quad C_g = C_g(L),\quad C_{\mathrm{BP}} = C_{\mathrm{BP}}(T,c_{\mathrm{BDG}}),$$

so $C_\Phi = C_\Phi(T,L,c_{\mathrm{BDG}})$. Setting

$$\lambda_* := (2\sqrt{C_\Phi})^{-1}\in(0,\tfrac12]$$

gives $C_\Phi \lambda^2 \le \frac{1}{4}$ for $\lambda \in [0, \lambda_*]$, so that (4.13) reads

$$\|\Phi_\lambda(\Theta^1) - \Phi_\lambda(\Theta^2)\|_{\mathbb{S}} \le \tfrac{1}{2}\,\|\Theta^1 - \Theta^2\|_{\mathbb{S}}, \qquad \lambda \in [0, \lambda_*].$$

By the Banach fixed-point theorem, Φ_λ has a unique fixed point $\Theta \in \mathbb{S}$. Evaluating Φ_λ at Θ itself,

$$\mu^\Theta = \mu, \qquad \bar{\Theta} = \Theta,$$

so (4.7)–(4.8) become the forward and backward equations of (4.1), which is solved by Θ.

Conversely. Let $\Theta \in \mathbb{S}$ solve (4.1) and evaluate Φ_λ at the trial Θ. Again $\mu^\Theta = \mu$, so (4.7) and the forward equation of (4.1) have the same right-hand side, which does not involve $\bar{X}$; hence $\bar{X} = X$. For the backward component set

$$M'_t := Y_t + \int_0^t \Big(\lambda f(s, \Theta_s, \mu_s) + q^f_s\Big)\,\mathrm{d}s,$$

so that

$$M'_T = \Xi, \qquad \mathrm{d}M'_t = Z_t\,\mathrm{d}W_t + \int_E U_t(e)\,\widetilde{N}(\mathrm{d}t, \mathrm{d}e).$$

Since $Z \in \mathcal{H}^2_W$ and $U \in \mathcal{H}^2_N$, M' is a true L^2-martingale, so

$$M'_t = \mathbb{E}[\Xi \mid \mathcal{F}_t] = M_t.$$

Uniqueness of the predictable representation forces $(\bar{Z}, \bar{U}) = (Z, U)$, and (4.8) then gives $\bar{Y} = Y$, so

$$\bar{\Theta} = \Theta.$$

Solutions are therefore exactly the fixed points, and the solution is unique; as q was arbitrary, $\mathsf{WP}(\lambda)$ holds on $[0, \lambda_*]$. □

4.3 Continuous perturbation stability

Lemma 4.3 (Continuous perturbation stability along the homotopy). *Let Assumptions 2.3–2.5 hold, fix $\underline{\lambda} \in (0, 1]$, and fix $\lambda \in [\underline{\lambda}, 1]$. Let Θ^1, Θ^2 solve (4.1) at parameter λ with $\mathcal{F}_0$-measurable initial data ξ^1_0, ξ^2_0 and forcings $q^k = (q^{b,k}, q^{\sigma,k}, q^{h,k}, q^{f,k}, q^{g,k}_T)$, $k = 1, 2$, where, for each k, X^k, Y^k are càdlàg adapted with $\mathbb{E}\int_0^T (|X^k_t|^2 + |Y^k_t|^2)\,\mathrm{d}t < \infty$ and $X^k_T \in L^2(\Omega, \mathcal{F}_T; \mathbb{R}^n)$, $Z^k \in \mathcal{H}^2_W$, $U^k \in \mathcal{H}^2_N$, and $\Theta^1 - \Theta^2 \in \mathbb{S}$. Then, writing $\delta\cdot = \cdot^1 - \cdot^2$,*

$$\begin{aligned}
&\mathbb{E}\sup_{t\le T}|\delta X_t|^2 + \mathbb{E}\sup_{t\le T}|\delta Y_t|^2 + \mathbb{E}\int_0^T \Big(|\delta Z_t|_F^2 + \|\delta U_t\|^2_{L^2(\nu)}\Big)\mathrm{d}t \\
&\qquad \le C_B\Big(\mathbb{E}|\delta\xi_0|^2 + \mathbb{E}|\delta q^g_T|^2 + \mathbb{E}\int_0^T \Big(|\delta q^b_t|^2 + |\delta q^\sigma_t|_F^2 + \|\delta q^h_t\|^2_{L^2(\nu)} + |\delta q^f_t|^2\Big)\mathrm{d}t\Big),
\end{aligned} \tag{4.14}$$

where C_B depends only on $T, L, \|G\|, c_G, \alpha, \beta_1, \beta_2, \underline{\lambda}$ and c_{BDG}.

Proof. The difference equations below involve the forcings only through δq, which we therefore write simply as q. We define

$$P := \mathbb{E}|\delta\xi_0|^2 + \mathbb{E}|q_T^g|^2 + \mathbb{E}\int_0^T \Big(|q_t^b|^2 + |q_t^\sigma|_F^2 + \|q_t^h\|_{L^2(\nu)}^2 + |q_t^f|^2\Big)\mathrm{d}t.$$

Set

$$\delta\psi_t := \psi(t, \Theta_t^1, \mu_t^1) - \psi(t, \Theta_t^2, \mu_t^2), \qquad \psi \in \{b, \sigma, h, f\},$$

where $\mu_t^k := \mathrm{Law}(\Theta_t^k)$; likewise δg_T, with δh evaluated at the predictable tuples $(\Theta_t^{k,-}, \mu_t^{k,-})$. For a.e. t each pair (Θ_t^k, μ_t^k) is a diagonal input in the sense of Definition 2.1: $\Theta_t^k \in L^2(\Omega, \mathcal{F}_t; \mathcal{H})$ for a.e. t by Fubini, and μ_t^k is the Borel law flow of Lemma A.1. The difference equations are

$$\mathrm{d}\,\delta X_t = (\lambda\delta b_t + q_t^b)\,\mathrm{d}t + (\lambda\delta\sigma_t + q_t^\sigma)\,\mathrm{d}W_t + \int_E (\lambda\delta h_t(e) + q_t^h(e))\,\widetilde{N}(\mathrm{d}t, \mathrm{d}e), \tag{4.15}$$

$$\mathrm{d}\,\delta Y_t = -(\lambda\delta f_t + q_t^f)\,\mathrm{d}t + \delta Z_t\,\mathrm{d}W_t + \int_E \delta U_t(e)\,\widetilde{N}(\mathrm{d}t, \mathrm{d}e), \qquad \delta Y_T = \lambda\,\delta g_T + q_T^g. \tag{4.16}$$

The five steps run as follows. Steps 1–2 pair $G\,\delta X$ against δY and insert Assumption 2.5: this controls the projections $G\delta X$ and $G^\top\delta Y, G^\top\delta Z, G^\top\delta U$, but only by the data P together with a free multiple of the full norm. Steps 3–4 record two a priori estimates, the forward one bounding δX through $(\delta Y, \delta Z, \delta U)$ and the backward one bounding $(\delta Y, \delta Z, \delta U)$ through δX; neither closes alone. Step 5 breaks the circle: full rank makes $G^\top$ injective when $n \geq m$ and G injective when $m \geq n$, so the coercive blocks present shed their projections, and which blocks are present decides whether the forward or the backward estimate enters first. The free multiple is then chosen small and absorbed.

Step 1 (G-energy identity). Both δX and δY jump only through their compensated Poisson integrals, with $\Delta\,\delta X_t = (\lambda\delta h_t + q_t^h)(e)$ and $\Delta\,\delta Y_t = \delta U_t(e)$ at each atom (t, e) of N; hence the jump covariation splits through $N = \widetilde{N} + \mathrm{d}t\,\nu$ as

$$\begin{aligned}
\sum_{t \leq T} \langle G\,\Delta\delta X_t,\, \Delta\delta Y_t\rangle &= \int_0^T\!\!\int_E \langle G(\lambda\delta h_t + q_t^h)(e),\, \delta U_t(e)\rangle\, N(\mathrm{d}t, \mathrm{d}e) \\
&= \int_0^T\!\!\int_E \langle G(\lambda\delta h_t + q_t^h)(e),\, \delta U_t(e)\rangle\, \widetilde{N}(\mathrm{d}t, \mathrm{d}e) \\
&\quad + \int_0^T\!\!\int_E \langle \lambda\delta h_t(e) + q_t^h(e),\, G^\top\delta U_t(e)\rangle\, \nu(\mathrm{d}e)\mathrm{d}t.
\end{aligned}$$

The continuous covariation of δX and δY, whose diffusions are $\lambda\delta\sigma_t + q_t^\sigma$ and δZ_t, contributes

$$\int_0^T \langle \lambda\delta\sigma_t + q_t^\sigma,\, G^\top\delta Z_t\rangle_F\,\mathrm{d}t.$$

Itô's product formula for $\langle G\,\delta X_t,\, \delta Y_t\rangle$, integrated over $[0, T]$ and using $\delta X_0 = \delta\xi_0$, then gives the pathwise identity

$$
\begin{aligned}
\langle G\delta X_T,\, \delta Y_T\rangle - \langle G\,\delta\xi_0,\, \delta Y_0\rangle = &\int_0^T \langle \lambda\delta b_t + q_t^b,\, G^\top \delta Y_{t-}\rangle\,\mathrm{d}t\\
&+ \int_0^T \langle \lambda\delta\sigma_t + q_t^\sigma,\, G^\top \delta Z_t\rangle_F\,\mathrm{d}t\\
&+ \int_0^T\int_E \langle \lambda\delta h_t(e) + q_t^h(e),\, G^\top \delta U_t(e)\rangle\,\nu(\mathrm{d}e)\mathrm{d}t\\
&+ \int_0^T \langle -\lambda\delta f_t - q_t^f,\, G\delta X_{t-}\rangle\,\mathrm{d}t\\
&+ \int_0^T \langle G\delta X_{t-},\, \delta Z_t\,\mathrm{d}W_t\rangle\\
&+ \int_0^T \langle G(\lambda\delta\sigma_t + q_t^\sigma)\,\mathrm{d}W_t,\, \delta Y_{t-}\rangle\\
&+ \int_0^T\int_E \langle G\delta X_{t-},\, \delta U_t(e)\rangle\,\widetilde{N}(\mathrm{d}t,\mathrm{d}e)\\
&+ \int_0^T\int_E \langle G(\lambda\delta h_t + q_t^h)(e),\, \delta Y_{t-}\rangle\,\widetilde{N}(\mathrm{d}t,\mathrm{d}e)\\
&+ \int_0^T\int_E \langle G(\lambda\delta h_t + q_t^h)(e),\, \delta U_t(e)\rangle\,\widetilde{N}(\mathrm{d}t,\mathrm{d}e).
\end{aligned}
\tag{4.17}
$$

Of the last five terms of (4.17), the first four are bounded by BDG and Cauchy–Schwarz, with (2.1) in the jump terms:

$$
\begin{cases}
\mathbb{E}\sup_{t\le T}\left|\int_0^t \langle G\delta X_{s-},\, \delta Z_s\,\mathrm{d}W_s\rangle\right|\\
\qquad\qquad \le c_{\mathrm{BDG}}\|G\|\Big(\mathbb{E}\sup_{t\le T}|\delta X_t|^2\Big)^{1/2}\Big(\mathbb{E}\int_0^T |\delta Z_t|_F^2\,\mathrm{d}t\Big)^{1/2},\\
\mathbb{E}\sup_{t\le T}\left|\int_0^t \langle G(\lambda\delta\sigma_s + q_s^\sigma)\,\mathrm{d}W_s,\, \delta Y_{s-}\rangle\right|\\
\qquad\qquad \le c_{\mathrm{BDG}}\|G\|\Big(\mathbb{E}\sup_{t\le T}|\delta Y_t|^2\Big)^{1/2}\Big(\mathbb{E}\int_0^T |\lambda\delta\sigma_t + q_t^\sigma|_F^2\,\mathrm{d}t\Big)^{1/2},\\
\mathbb{E}\sup_{t\le T}\left|\int_0^t\int_E \langle G\delta X_{s-},\, \delta U_s(e)\rangle\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e)\right|\\
\qquad\qquad \le c_{\mathrm{BDG}}\|G\|\Big(\mathbb{E}\sup_{t\le T}|\delta X_t|^2\Big)^{1/2}\Big(\mathbb{E}\int_0^T \|\delta U_t\|_{L^2(\nu)}^2\,\mathrm{d}t\Big)^{1/2},\\
\mathbb{E}\sup_{t\le T}\left|\int_0^t\int_E \langle G(\lambda\delta h_s + q_s^h)(e),\, \delta Y_{s-}\rangle\,\widetilde{N}(\mathrm{d}s,\mathrm{d}e)\right|\\
\qquad\qquad \le c_{\mathrm{BDG}}\|G\|\Big(\mathbb{E}\sup_{t\le T}|\delta Y_t|^2\Big)^{1/2}\Big(\mathbb{E}\int_0^T \|\lambda\delta h_t + q_t^h\|_{L^2(\nu)}^2\,\mathrm{d}t\Big)^{1/2}.
\end{cases}
\tag{4.18}
$$

Since $\Theta^1 - \Theta^2 \in \mathbb{S}$,

$$
\mathbb{E}\sup_{t\le T}|\delta X_t|^2 + \mathbb{E}\sup_{t\le T}|\delta Y_t|^2 + \mathbb{E}\int_0^T \Big(|\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t = \|\Theta^1 - \Theta^2\|_{\mathbb{S}}^2 < \infty,
$$

and by the computation of (4.9), with the forcings square-integrable,

$$\mathbb{E}\int_0^T \Big(|\lambda\delta\sigma_t + q_t^\sigma|_F^2 + \|\lambda\delta h_t + q_t^h\|_{L^2(\nu)}^2\Big)\mathrm{d}t < \infty.$$

The four suprema in (4.18) are therefore integrable, so the supremum criterion applies to those four local martingales. The fifth has predictable integrand, and by Cauchy–Schwarz

$$\begin{aligned}\mathbb{E}\int_0^T\int_E \Big|\langle \lambda\delta h_t(e) + q_t^h(e),\, G^\top\delta U_t(e)\rangle\Big|\,\nu(\mathrm{d}e)\mathrm{d}t&\\ &\le \|G\|\Big(\mathbb{E}\int_0^T \|\lambda\delta h_t + q_t^h\|_{L^2(\nu)}^2\mathrm{d}t\Big)^{1/2}\Big(\mathbb{E}\int_0^T \|\delta U_t\|_{L^2(\nu)}^2\mathrm{d}t\Big)^{1/2}\\ &< \infty.\end{aligned}$$

Hence (2.2) applies, and all five terms have zero expectation.

Taking expectations and dropping the left limits in the drift multipliers (the dt-integral convention above) reduces (4.17) to

$$\begin{aligned}\mathbb{E}\langle G\delta X_T,\, \delta Y_T\rangle - \mathbb{E}\langle G\,\delta\xi_0,\, \delta Y_0\rangle = \mathbb{E}\int_0^T \Big[&\langle \lambda\delta b_t + q_t^b,\, G^\top\delta Y_t\rangle + \langle \lambda\delta\sigma_t + q_t^\sigma,\, G^\top\delta Z_t\rangle_F\\ &+\int_E \langle \lambda\delta h_t(e) + q_t^h(e),\, G^\top\delta U_t(e)\rangle\nu(\mathrm{d}e) + \langle -\lambda\delta f_t - q_t^f,\, G\delta X_t\rangle\Big]\mathrm{d}t. \end{aligned}\tag{4.19}$$

Step 2 (insertion of monotonicity). In the h-channel δh_t is evaluated at the predictable tuples $(\Theta_t^{k,-}, \mu_t^{k,-})$, which agree with (Θ_t^k, μ_t^k) for dt-a.e. t by Lemma A.1. Assumption 2.5, applied at the common input pair $(\Theta_t^1, \mu_t^1), (\Theta_t^2, \mu_t^2)$, therefore bounds the unperturbed part of the integrand of (4.19):

$$\begin{aligned}&\lambda\mathbb{E}\Big[\langle \delta b_t,\, G^\top\delta Y_t\rangle + \langle \delta\sigma_t,\, G^\top\delta Z_t\rangle_F + \int_E \langle \delta h_t(e),\, G^\top\delta U_t(e)\rangle\nu(\mathrm{d}e) + \langle -\delta f_t,\, G\delta X_t\rangle\Big]\\ &\quad\le -\lambda\beta_1\mathbb{E}\Big[|G^\top\delta Y_t|^2 + |G^\top\delta Z_t|_F^2 + \|G^\top\delta U_t\|_{L^2(\nu)}^2\Big] - \lambda\beta_2\mathbb{E}|G\delta X_t|^2.\end{aligned}\tag{4.20}$$

The terminal side of (4.19) is bounded by the terminal monotonicity of Assumption 2.5:

$$\begin{aligned}\mathbb{E}\langle G\delta X_T,\, \delta Y_T\rangle &= \lambda\mathbb{E}\langle G\delta X_T,\, \delta g_T\rangle + \mathbb{E}\langle G\delta X_T,\, q_T^g\rangle\\ &\ge \lambda\alpha\mathbb{E}|G\delta X_T|^2 - |\mathbb{E}\langle G\delta X_T,\, q_T^g\rangle|.\end{aligned}\tag{4.21}$$

Recall

$$P = \mathbb{E}|\delta\xi_0|^2 + \mathbb{E}|q_T^g|^2 + \mathbb{E}\int_0^T \Big(|q_t^b|^2 + |q_t^\sigma|_F^2 + \|q_t^h\|_{L^2(\nu)}^2 + |q_t^f|^2\Big)\mathrm{d}t,$$

and set

$$\begin{aligned}\mathcal{N}' &:= \mathbb{E}|\delta X_T|^2 + \mathbb{E}|\delta Y_0|^2 + \mathbb{E}\int_0^T \Big(|\delta X_t|^2 + |\delta Y_t|^2 + |\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t,\\ \mathcal{K} &:= \eta\mathcal{N}' + \tfrac{\|G\|^2}{4\eta}\,P, \qquad \eta > 0.\end{aligned}$$

Inserting (4.20) and (4.21) into (4.19) gives

$$\begin{aligned}
&\lambda\alpha\mathbb{E}|G\delta X_T|^2 - |\mathbb{E}\langle G\delta X_T,\, q_T^g\rangle| - \mathbb{E}\langle G\,\delta\xi_0,\, \delta Y_0\rangle \\
&\quad \le -\lambda\beta_1\mathbb{E}\int_0^T \Big[|G^\top\delta Y_t|^2 + |G^\top\delta Z_t|_F^2 + \|G^\top\delta U_t\|_{L^2(\nu)}^2\Big]\mathrm{d}t - \lambda\beta_2\mathbb{E}\int_0^T |G\delta X_t|^2\mathrm{d}t \\
&\quad + \mathbb{E}\int_0^T\Big(\langle q_t^b,\, G^\top\delta Y_t\rangle + \langle q_t^\sigma,\, G^\top\delta Z_t\rangle_F + \langle -q_t^f,\, G\delta X_t\rangle \\
&\qquad\qquad + \int_E \langle q_t^h(e),\, G^\top\delta U_t(e)\rangle\,\nu(\mathrm{d}e)\Big)\mathrm{d}t.
\end{aligned}$$

Rearranging, and applying Young's inequality with $|G^\top v| \le \|G\|\,|v|$ and $|Gv| \le \|G\|\,|v|$, gives

$$\begin{aligned}
&\lambda\alpha\mathbb{E}|G\delta X_T|^2 + \lambda\beta_2\mathbb{E}\int_0^T |G\delta X_t|^2\mathrm{d}t \\
&\quad + \lambda\beta_1\mathbb{E}\int_0^T\Big[|G^\top\delta Y_t|^2 + |G^\top\delta Z_t|_F^2 + \|G^\top\delta U_t\|_{L^2(\nu)}^2\Big]\mathrm{d}t \\
&\qquad \le |\mathbb{E}\langle G\,\delta\xi_0,\, \delta Y_0\rangle| + |\mathbb{E}\langle G\delta X_T,\, q_T^g\rangle| + \mathbb{E}\int_0^T\Big(|\langle q^b,\, G^\top\delta Y\rangle| + |\langle q^\sigma,\, G^\top\delta Z\rangle_F|\Big)\mathrm{d}t \\
&\qquad + \mathbb{E}\int_0^T\Big(\Big|\int_E\langle q^h(e),\, G^\top\delta U(e)\rangle\,\nu(\mathrm{d}e)\Big| + |\langle q^f,\, G\delta X\rangle|\Big)\mathrm{d}t \\
&\qquad \le \mathcal{K}.
\end{aligned} \tag{4.22}$$

Each coercive block present on the left of (4.22) is therefore at most $\mathcal{K}$.

Step 3 (forward estimate). Fix $t \le T$ and integrate (4.15) over $[0,s]$, $s \le t$, with $\delta X_0 = \delta\xi_0$. Squaring, taking the supremum over s and expectations, Cauchy–Schwarz on the drift, BDG on the two martingales and Assumption 2.3 give

$$\begin{aligned}
\mathbb{E}\sup_{s\le t}|\delta X_s|^2 &\le C\Big(\mathbb{E}|\delta\xi_0|^2 + \mathbb{E}\int_0^t\Big(|\delta b_s|^2 + |\delta\sigma_s|_F^2 + \|\delta h_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s + P\Big) \\
&\le C\Big(\mathbb{E}|\delta\xi_0|^2 + \mathbb{E}\int_0^t\Big(|\delta X_s|^2 + |\delta Y_s|^2 + |\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s + P\Big) \\
&\le C\Big(\int_0^t \mathbb{E}\sup_{r\le s}|\delta X_r|^2\,\mathrm{d}s + \mathbb{E}\int_0^T\Big(|\delta Y_s|^2 + |\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s + P\Big).
\end{aligned}$$

The map $t \mapsto \mathbb{E}\sup_{s\le t}|\delta X_s|^2$ is nondecreasing and finite, since $\Theta^1 - \Theta^2 \in \mathbb{S}$, so Grönwall gives

$$\mathbb{E}\sup_{t\le T}|\delta X_t|^2 \le C\Big(\mathbb{E}\int_0^T\Big(|\delta Y_t|^2 + |\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t + P\Big). \tag{4.23}$$

Step 4 (backward estimate). Itô's formula applied to $e^{\kappa t}|\delta Y_t|^2$ gives, for $\kappa > 0$,

$$\begin{aligned}
&\mathbb{E}\,e^{\kappa t}|\delta Y_t|^2 + \mathbb{E}\int_t^T e^{\kappa s}\Big(\kappa|\delta Y_s|^2 + |\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s \\
&\qquad \le \mathbb{E}\,e^{\kappa T}|\lambda\delta g_T + q_T^g|^2 + 2\,\mathbb{E}\int_t^T e^{\kappa s}\langle\delta Y_s,\, \lambda\delta f_s + q_s^f\rangle\mathrm{d}s.
\end{aligned} \tag{4.24}$$

By Assumption 2.3, the coupling bound (2.8), and Young's inequality, for every $\varepsilon > 0$,

$$2\,\mathbb{E}\langle \delta Y_s,\, \lambda\delta f_s + q_s^f\rangle \le \frac{C}{\varepsilon}\,\mathbb{E}|\delta Y_s|^2 + \varepsilon\,\mathbb{E}\Big(|\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big) + \frac{C}{\varepsilon}\,\mathbb{E}\Big(|\delta X_s|^2 + |q_s^f|^2\Big), \quad (4.25)$$

with C independent of ε.

Insert (4.25) in (4.24):

$$\begin{aligned}\mathbb{E}\,e^{\kappa t}|\delta Y_t|^2 + \mathbb{E}\int_t^T e^{\kappa s}\Big[\Big(\kappa - \tfrac{C}{\varepsilon}\Big)|\delta Y_s|^2 + (1-\varepsilon)\Big(|\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\Big]\mathrm{d}s \\ \le \mathbb{E}\,e^{\kappa T}|\lambda\delta g_T + q_T^g|^2 + \tfrac{C}{\varepsilon}\,\mathbb{E}\int_t^T e^{\kappa s}\Big(|\delta X_s|^2 + |q_s^f|^2\Big)\mathrm{d}s.\end{aligned}$$

Take $\varepsilon < 1$ and $\kappa > C/\varepsilon$. The δY-integral on the left is then nonnegative and may be dropped, and the weights satisfy $1 \le e^{\kappa s} \le e^{\kappa T}$ on $[0,T]$, so

$$\mathbb{E}|\delta Y_t|^2 + \mathbb{E}\int_t^T\Big(|\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s \le C\Big(\mathbb{E}|\lambda\delta g_T + q_T^g|^2 + \mathbb{E}\int_t^T\Big(|\delta X_s|^2 + |q_s^f|^2\Big)\mathrm{d}s\Big).$$

With $\mathbb{E}|\delta g_T|^2 \le C\,\mathbb{E}|\delta X_T|^2$, the forcings in P, and $\int_t^T \le \int_0^T$ on the right, the bound no longer depends on t. Taking the supremum over t in the first term on the left and $t=0$ in the second, there remains

$$\sup_{t\le T}\mathbb{E}|\delta Y_t|^2 + \mathbb{E}\int_0^T\Big(|\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t \le C\Big(\mathbb{E}|\delta X_T|^2 + \mathbb{E}\int_0^T|\delta X_t|^2\mathrm{d}t + P\Big). \quad (4.26)$$

Writing (4.16) forward from δY_0, BDG, Cauchy–Schwarz, Assumption 2.3 with the coupling bound (2.8), and (4.26) give

$$\begin{aligned}\mathbb{E}\sup_{t\le T}|\delta Y_t|^2 &\le C\Big(\mathbb{E}|\delta Y_0|^2 + \mathbb{E}\int_0^T|\lambda\,\delta f_s + q_s^f|^2\mathrm{d}s + \mathbb{E}\int_0^T\Big(|\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s\Big) \\ &\le C\Big(\mathbb{E}|\delta Y_0|^2 + \mathbb{E}\int_0^T\Big(|\delta X_s|^2 + |\delta Y_s|^2 + |\delta Z_s|_F^2 + \|\delta U_s\|_{L^2(\nu)}^2\Big)\mathrm{d}s + P\Big) \\ &\le C\Big(\mathbb{E}|\delta X_T|^2 + \mathbb{E}\int_0^T|\delta X_t|^2\mathrm{d}t + P\Big).\end{aligned}$$

The first term of (4.26) may therefore be replaced by $\mathbb{E}\sup_{t\le T}|\delta Y_t|^2$:

$$\mathbb{E}\sup_{t\le T}|\delta Y_t|^2 + \mathbb{E}\int_0^T\Big(|\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t \le C\Big(\mathbb{E}|\delta X_T|^2 + \mathbb{E}\int_0^T|\delta X_t|^2\mathrm{d}t + P\Big). \quad (4.27)$$

Step 5 (rank-case closure, then absorb). Write

$$\mathcal{N} := \mathbb{E}\sup_{t\le T}|\delta X_t|^2 + \mathbb{E}\sup_{t\le T}|\delta Y_t|^2 + \mathbb{E}\int_0^T\Big(|\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t$$

so that (4.14) is the claim $\mathcal{N} \le C_B P$. Then $\mathcal{N}' \le (1+2T)\mathcal{N}$, and by (4.22) each coercive block on its left is at most $\mathcal{K}$.

Case A ($n \geq m$; active when $m < n$, where $\beta_1 > 0$). As $G^\top$ is injective, Assumption 2.4 gives

$$|\delta Y|^2 \leq c_G |G^\top \delta Y|^2, \qquad |\delta Z|_F^2 \leq c_G |G^\top \delta Z|_F^2, \qquad \|\delta U\|_{L^2(\nu)}^2 \leq c_G \|G^\top \delta U\|_{L^2(\nu)}^2.$$

With $\beta_1 > 0$ and $\lambda \geq \underline{\lambda}$, the β_1-block of (4.22) yields

$$\mathbb{E}\int_0^T \Big(|\delta Y_t|^2 + |\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t \leq \Big(c_G/(\underline{\lambda}\beta_1)\Big)\mathcal{K}.$$

Inserting this in (4.23), and the result in (4.27),

$$\begin{aligned}\mathbb{E}\sup_{t\leq T}|\delta X_t|^2 &\leq C(\mathcal{K}+P),\\ \mathbb{E}\sup_{t\leq T}|\delta Y_t|^2 + \mathbb{E}\int_0^T \Big(|\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t &\leq C(\mathcal{K}+P),\end{aligned}$$

so $\mathcal{N} \leq C(\mathcal{K}+P)$.

Case B ($m \geq n$; active when $m > n$, where $\alpha, \beta_2 > 0$). As G is injective, Assumption 2.4 gives

$$|\delta X|^2 \leq c_G |G\delta X|^2.$$

With $\alpha, \beta_2 > 0$ and $\lambda \geq \underline{\lambda}$, the α, β_2-blocks of (4.22) yield

$$\mathbb{E}|\delta X_T|^2 + \mathbb{E}\int_0^T |\delta X_t|^2 \mathrm{d}t \leq \Big(2c_G/(\underline{\lambda}\min\{\alpha, \beta_2\})\Big)\mathcal{K}.$$

Inserting this in (4.27), and the result in (4.23),

$$\begin{aligned}\mathbb{E}\sup_{t\leq T}|\delta Y_t|^2 + \mathbb{E}\int_0^T \Big(|\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t &\leq C(\mathcal{K}+P),\\ \mathbb{E}\sup_{t\leq T}|\delta X_t|^2 &\leq C(\mathcal{K}+P),\end{aligned}$$

so again $\mathcal{N} \leq C(\mathcal{K}+P)$.

When $m = n$, both $G, G^\top$ are injective, so Case A is available whenever $\beta_1 > 0$ and Case B whenever $\alpha, \beta_2 > 0$; Assumption 2.5 provides at least one of the two, and when both are available we use whichever gives the smaller bound. In every case

$$\mathcal{N} \leq C\Big(\eta\mathcal{N}' + \tfrac{\|G\|^2}{4\eta}P + P\Big) \leq C\eta(1+2T)\mathcal{N} + C\,P;$$

choosing η small enough that $C\eta(1+2T) \leq \frac{1}{2}$ absorbs the $\mathcal{N}$-term (finite, since $\Theta^1 - \Theta^2 \in \mathbb{S}$) and gives $\mathcal{N} \leq C\,P$, which is (4.14) with $C_B := C$. □

4.4 Stability

Proof of Theorem 3.1. Take the system-1 coefficients $(b^1, \sigma^1, h^1, f^1, g^1)$ as the baseline and read Θ^2 as a forced solution of that system: Θ^2 solves (4.1) at $\lambda = 1$ with

$$q^{\psi,2} := -(\delta\psi), \qquad \psi \in \{b, \sigma, h, f\}, \qquad q_T^{g,2} := -\widehat{\delta g}_T,$$

while Θ^1 solves it with zero forcing.

These form a square-integrable forcing for (4.1): as differences of the coefficient processes (2.5), the $q^{\psi,2}$ inherit their measurability, and they are square-integrable by (4.2) applied to each system with its own constants; $q_T^{g,2} \in L^2(\Omega, \mathcal{F}_T; \mathbb{R}^m)$ by (2.6) with $X_T^2 \in L^2$.

Since $q^1 = 0$, the difference $\delta q = q^1 - q^2$ has $\delta q^\psi = (\delta\psi)$ and $\delta q_T^g = \widehat{\delta g}_T$, so Lemma 4.3 at parameter $\lambda = 1$ and lower bound $\underline{\lambda} = 1$, applied to system 1, gives

$$\begin{aligned}\mathbb{E}\sup_{t\le T}|\delta X_t|^2 + \mathbb{E}\sup_{t\le T}|\delta Y_t|^2 + \mathbb{E}\int_0^T \Big(|\delta Z_t|_F^2 + \|\delta U_t\|_{L^2(\nu)}^2\Big)\mathrm{d}t \\ \le C_B\Big(\mathbb{E}|\delta\xi_0|^2 + \mathbb{E}\Big|\widehat{\delta g}_T\Big|^2 \\ + \mathbb{E}\int_0^T \Big(|(\delta b)(t)|^2 + |(\delta\sigma)(t)|_F^2 + \|(\delta h)(t)\|_{L^2(\nu)}^2 + |(\delta f)(t)|^2\Big)\mathrm{d}t\Big),\end{aligned}$$

which is (3.1) with $C_{\mathrm{stab}} := C_B$. ☐

4.5 The continuation step

Taking identical sources and the common initial datum in Lemma 4.3 (at lower bound $\underline{\lambda} = \lambda_*$) yields uniqueness at each $\lambda \in [\lambda_*, 1]$; Lemma 4.2 gives it on $[0, \lambda_*]$. It remains to prove existence up to $\lambda = 1$.

Lemma 4.4 (Continuation)**.** *Let Assumptions 2.2–2.5 hold, and let λ_* be the small-coupling threshold of Lemma 4.2. There is $\zeta_0 > 0$, depending only on T, L, $\|G\|$, c_G, α, β_1, β_2, λ_* and c_{BDG}, such that: if $\lambda_0 \in [\lambda_*, 1]$ and $\mathsf{WP}(\lambda_0)$ holds, then $\mathsf{WP}(\lambda_0 + \zeta)$ holds for every $\zeta \in [0, \zeta_0]$ with $\lambda_0 + \zeta \le 1$.*

Proof. Fix λ_0 with $\mathsf{WP}(\lambda_0)$, set $\lambda_1 = \lambda_0 + \zeta \le 1$, and fix a forcing q for the λ_1-system. For a trial $\Theta \in \mathbb{S}$ define the augmented forcing

$$R^\psi(\Theta) := q^\psi + \zeta\,\psi(\cdot, \Theta, \mu^\Theta), \quad \psi \in \{b, \sigma, h, f\}, \qquad R_T^g(\Theta) := q_T^g + \zeta\, g(X_T, \mathrm{Law}(X_T)).$$

Then $R(\Theta)$ is again a square-integrable forcing for (4.1). Its integrability follows from (4.2) and the Lipschitz/coupling bounds, and its h-channel is evaluated at the predictable tuple $(\Theta^-, \mu^{\Theta,-})$, hence predictable. Since $\mathsf{WP}(\lambda_0)$ holds, let $\Phi(\Theta) := \bar{\Theta}$ be the unique solution in $\mathbb{S}$ of the forced λ_0-system with forcing $R(\Theta)$.

For Θ^1, Θ^2, the outputs $\Phi(\Theta^k)$ solve the same λ_0-system with sources $R(\Theta^1), R(\Theta^2)$; q cancels in the difference, leaving

$$R^\psi(\Theta^1) - R^\psi(\Theta^2) = \zeta[\psi(\cdot, \Theta^1, \mu^{\Theta^1}) - \psi(\cdot, \Theta^2, \mu^{\Theta^2})].$$

By Lemma 4.3 at parameter λ_0 and lower bound $\underline{\lambda} = \lambda_*$, and the Lipschitz bound of Assumption 2.3,

$$\begin{aligned}\|\Phi(\Theta^1) - \Phi(\Theta^2)\|_{\mathbb{S}}^2 &\leq C_B \sum_\psi \mathbb{E}\int_0^T |R^\psi(\Theta^1) - R^\psi(\Theta^2)|^2 \mathrm{d}t + C_B\, \mathbb{E}|R_T^g(\Theta^1) - R_T^g(\Theta^2)|^2 \\ &\leq C_B\, C\, \zeta^2\, \|\Theta^1 - \Theta^2\|_{\mathbb{S}}^2, \qquad C = C(T, L).\end{aligned}$$

Put $\zeta_0 := \min\{1, (2\sqrt{C_B C})^{-1}\}$, so that for $\zeta \leq \zeta_0$

$$\|\Phi(\Theta^1) - \Phi(\Theta^2)\|_{\mathbb{S}} \leq \tfrac{1}{2}\, \|\Theta^1 - \Theta^2\|_{\mathbb{S}}.$$

By the Banach fixed-point theorem Φ has a unique fixed point Θ, at which $R^\psi(\Theta) = q^\psi + \zeta\, \psi(\cdot, \Theta, \mu)$. Since $\lambda_0 + \zeta = \lambda_1$,

$$\begin{aligned}\lambda_0\, \psi(\cdot, \Theta, \mu) + R^\psi(\Theta) &= \lambda_1\, \psi(\cdot, \Theta, \mu) + q^\psi, \qquad \psi \in \{b, \sigma, h, f\}, \\ \lambda_0\, g(X_T, \mathrm{Law}(X_T)) + R_T^g(\Theta) &= \lambda_1\, g(X_T, \mathrm{Law}(X_T)) + q_T^g,\end{aligned}$$

so Θ solves the forced λ_1-system (4.1) with forcing q.

Any solution of the λ_1-system is a fixed point of the same Φ, which has only one, so the solution is unique. Hence $\mathsf{WP}(\lambda_1)$, and ζ_0 is independent of λ_0 and q. □

4.6 Proof of Theorem 3.5

Proof. By Lemma 4.2, $\mathsf{WP}(\lambda)$ holds for $\lambda \in [0, \lambda_*]$. With ζ_0 from Lemma 4.4, set

$$\lambda_k := \min\{\lambda_* + k\zeta_0,\, 1\}, \qquad k \geq 0,$$

so that $\lambda_0 = \lambda_*$, $\lambda_{k+1} - \lambda_k \leq \zeta_0$ and $\lambda_k = 1$ once $k \geq \lceil (1 - \lambda_*)/\zeta_0 \rceil$. Lemma 4.4 carries $\mathsf{WP}(\lambda_k)$ to $\mathsf{WP}(\lambda_{k+1})$ at every step, so $\mathsf{WP}(1)$ holds; taking $q \equiv 0$ there gives a unique solution of (1.1) in $\mathbb{S}$. □

5 Examples

Theorem 3.5 permits broad law dependence; the restrictive hypothesis is the dissipativity of Assumption 2.5, which in applications is the one to verify and which cannot be traded for Lipschitz continuity alone. The jump system of [14, Example 3.2] is linear, hence

Lipschitz, and is shown there to admit no adapted solution on the horizon $T = 3\pi/4$.

The examples illustrate both the scope of the law dependence and the role of the dissipativity assumption. Example 5.1 exhibits two dissipative models meeting every assumption exactly, both at arbitrary jump activity: (i) a law-free affine base, and (ii) mean-field coefficients. Examples 5.2 and 5.3 recover the Brownian [3] and the expectation-functional [14] subclasses as special cases. Example 5.4 then carries the law dependence beyond those subclasses, to genuinely nonlinear functionals of the full tuple and of the jump integrand, and Example 5.5, a non-perturbative full U-law model, is a mean-field dealer market in which the law of the $L^2(\nu)$-valued jump integrand arises as a population law on valuation curves and enters the jump response through a monotone crowding operator, so that well-posedness holds at every interaction strength.

Example 5.1 (Dissipative models).

(i) A law-free affine base. Take

$$n = m = d = 1, \qquad G = 1, \qquad \nu \ \sigma\text{-finite}, \qquad \xi_0 \in L^2(\Omega, \mathcal{F}_0; \mathbb{R}^n),$$
$$\sigma_c \in \mathbb{R}, \qquad h_c \in L^2(\nu), \qquad \beta_1, \beta_2 > 0.$$

Consider an agent who corrects an exogenous noise baseline through square-integrable predictable controls (a^b, a^σ, a^h), the jump control a^h valued in $L^2(\nu)$,

$$\mathrm{d}X_t = a_t^b \,\mathrm{d}t + (\sigma_c + a_t^\sigma)\,\mathrm{d}W_t + \int_E \Big(h_c(e) + a_t^h(e)\Big)\,\widetilde{N}(\mathrm{d}t, \mathrm{d}e), \qquad X_0 = \xi_0,$$

at quadratic cost

$$\mathbb{E}\Big[\tfrac{1}{2}|X_T|^2 + \int_0^T \Big(\tfrac{\beta_2}{2}|X_t|^2 + \tfrac{1}{2\beta_1}\Big(|a_t^b|^2 + |a_t^\sigma|^2 + \|a_t^h\|^2_{L^2(\nu)}\Big)\Big)\mathrm{d}t\Big].$$

The Hamiltonian

$$H = a^b y + (\sigma_c + a^\sigma)\, z + \int_E \Big(h_c(e) + a^h(e)\Big) u(e)\,\nu(\mathrm{d}e)$$
$$+ \tfrac{\beta_2}{2}|x|^2 + \tfrac{1}{2\beta_1}\Big(|a^b|^2 + |a^\sigma|^2 + \|a^h\|^2_{L^2(\nu)}\Big)$$

is strictly convex in the controls, with pointwise minimiser

$$a^{b*} = -\beta_1 y, \qquad a^{\sigma*} = -\beta_1 z, \qquad a^{h*}(e) = -\beta_1 u(e),$$

so by the stochastic maximum principle for jump diffusions [20] the Pontryagin optimality system consists of the closed-loop state dynamics and the adjoint equation with driver $\partial_x H = \beta_2 x$ and terminal value X_T; this is (1.1) with the affine coefficients

$$b = -\beta_1 y, \qquad \sigma = \sigma_c - \beta_1 z, \qquad h(e) = h_c(e) - \beta_1 u(e),$$
$$f = \beta_2 x, \qquad g(x, \rho) = x.$$

Thus β_1 is the reciprocal of the quadratic actuation cost, β_2 the running state weight, the terminal weight is 1, and σ_c, h_c are the uncontrolled baseline. With these coefficients (1.1) reads

$$\begin{cases} X_t = \xi_0 - \beta_1 \int_0^t Y_s \, \mathrm{d}s + \int_0^t (\sigma_c - \beta_1 Z_s) \, \mathrm{d}W_s + \int_0^t \int_E (h_c(e) - \beta_1 U_s(e)) \, \widetilde{N}(\mathrm{d}s, \mathrm{d}e), \\ Y_t = X_T + \beta_2 \int_t^T X_s \, \mathrm{d}s - \int_t^T Z_s \, \mathrm{d}W_s - \int_t^T \int_E U_s(e) \, \widetilde{N}(\mathrm{d}s, \mathrm{d}e). \end{cases}$$

The coefficients are affine, hence Lipschitz, and their origin values, the constants σ_c and $h_c \in L^2(\nu)$, are square-integrable in time over the finite horizon, so Assumptions 2.2 and 2.3 hold with $L = \max\{\beta_1, \beta_2, 1\}$, where the 1 covers the terminal map g. The matrix $G = 1$ is invertible, so Assumption 2.4 holds with $c_G = 1$. Since $m = n$, both rank branches of Assumption 2.5 are available.

Substituting the coefficients, the interior and terminal pairings of (2.7) evaluate exactly to

$$\begin{aligned} &\mathbb{E}\Big[\langle \delta b, \, \delta Y \rangle + \langle \delta\sigma, \, \delta Z \rangle + \langle -\delta f, \, \delta X \rangle + {\textstyle\int_E} \langle \delta h(e), \, \delta U(e) \rangle \nu(\mathrm{d}e)\Big] \\ &\qquad = -\beta_1 \mathbb{E}\Big[|\delta Y|^2 + |\delta Z|_F^2 + \|\delta U\|_{L^2(\nu)}^2\Big] - \beta_2 \mathbb{E}|\delta X|^2, \\ &\mathbb{E}\langle \delta g_T, \, \delta X_T \rangle = \mathbb{E}|\delta X_T|^2. \end{aligned}$$

Assumption 2.5 holds with equality and $\alpha = 1$, so Theorem 3.5 yields well-posedness.

The identity is sharp for the constants β_1, β_2, so by Remark 2.11 it carries any margin $0 < c_{\mathrm{diss}} < \min\{\beta_1, \beta_2\}$, unweighted here because $G = 1$.

By Remark 2.10, *every* instance with $\beta_1 > 0$ must carry strictly U-dissipative jump feedback. The affine coefficient $h(e) = h_c(e) - \beta_1 u(e)$ realises that dissipativity with equality,

$$\int_E \langle \delta h(e), \, \delta U(e) \rangle \, \nu(\mathrm{d}e) = -\beta_1 \|\delta U\|_{L^2(\nu)}^2.$$

The baseline also realises the example of Proposition 3.8: replacing its drift $-\beta_1 y$ by (3.6) leaves (2.7) intact with constants $(\beta_1 - \varepsilon, \beta_2)$, so Theorem 3.5 continues to apply.

(ii) Mean-field terms at arbitrary jump activity. Write $\theta = (x, y, z, u)$ for a point of $\mathcal{H}$ and $\pi_\Psi(\theta)$ for its component indexed by $\Psi \in \{X, Y, Z, U\}$. The marginal means of μ and the barycentre of a terminal law ρ are

$$m_\Psi(\mu) := \int_{\mathcal{H}} \pi_\Psi(\theta) \, \mu(\mathrm{d}\theta), \qquad \mathfrak{b}(\rho) := \int_{\mathbb{R}^n} x \, \rho(\mathrm{d}x).$$

Take

$$\begin{aligned} &b = -2y - m_Y(\mu), && \sigma = -2z - m_Z(\mu), && h(e) = -2u(e) - m_U(\mu)(e), \\ &f = 2x + m_X(\mu), && g(x, \rho) = 2x + \mathfrak{b}(\rho). \end{aligned}$$

Keep $n = m = d = 1$ and $G = 1$ from (i). Set

$$\delta m_\Psi := m_\Psi(\mu) - m_\Psi(\mu'), \qquad \delta m := \Big(\delta m_X, \delta m_Y, \delta m_Z, \delta m_U\Big) = \int_{\mathcal{H}} \theta \,\mathrm{d}\mu - \int_{\mathcal{H}} \theta \,\mathrm{d}\mu',$$

so that δm is a single element of $\mathcal{H}$.

Both δm and $\mathfrak{b}(\rho)$ are integrals of the identity map, which is 1-Lipschitz, so (2.10) applies on $\mathcal{P}_2(\mathcal{H})$ and on $\mathcal{P}_2(\mathbb{R}^n)$. Cauchy–Schwarz across the four coordinates of $\mathcal{H}$ then splits the first bound into its components:

$$|\delta m|_{\mathcal{H}} \le W_2(\mu, \mu'), \qquad |\mathfrak{b}(\rho) - \mathfrak{b}(\rho')| \le W_2(\rho, \rho'),$$
$$|\delta m_X| + |\delta m_Y| + |\delta m_Z|_F + \|\delta m_U\|_{L^2(\nu)} \le 2\, |\delta m|_{\mathcal{H}} \le 2\, W_2(\mu, \mu').$$

Thus Assumption 2.3 holds with $L = 2$. The coefficients b, σ, h, f vanish at $(0, \delta_0)$, so Assumption 2.2 is immediate.

Along a diagonal pair the measure argument is the law of the realised tuple, so $\mathbb{E}\,\delta\Psi$ is deterministic and, for $\Psi \in \{X, Y, Z, U\}$,

$$m_\Psi(\mathrm{Law}(\Theta)) = \mathbb{E}\Psi, \qquad \delta m_\Psi = \mathbb{E}\,\delta\Psi, \qquad \mathbb{E}\langle \mathbb{E}\,\delta\Psi,\, \delta\Psi\rangle = |\mathbb{E}\,\delta\Psi|^2.$$

Substituting the coefficients and applying these identities, each channel contributes its own mean square,

$$\begin{aligned}
\mathbb{E}\langle -\delta f,\, \delta X\rangle &= \mathbb{E}\langle -2\,\delta X - \mathbb{E}\,\delta X,\, \delta X\rangle = -2\,\mathbb{E}|\delta X|^2 - |\mathbb{E}\delta X|^2,\\
\mathbb{E}\langle \delta b,\, \delta Y\rangle &= \mathbb{E}\langle -2\,\delta Y - \mathbb{E}\,\delta Y,\, \delta Y\rangle = -2\,\mathbb{E}|\delta Y|^2 - |\mathbb{E}\delta Y|^2,\\
\mathbb{E}\langle \delta \sigma,\, \delta Z\rangle &= \mathbb{E}\langle -2\,\delta Z - \mathbb{E}\,\delta Z,\, \delta Z\rangle = -2\,\mathbb{E}|\delta Z|_F^2 - |\mathbb{E}\delta Z|_F^2,\\
\mathbb{E}\int_E \langle \delta h(e),\, \delta U(e)\rangle\, \nu(\mathrm{d}e) &= \mathbb{E}\langle -2\,\delta U - \mathbb{E}\,\delta U,\, \delta U\rangle_{L^2(\nu)} = -2\,\mathbb{E}\|\delta U\|_{L^2(\nu)}^2 - \|\mathbb{E}\delta U\|_{L^2(\nu)}^2.
\end{aligned}$$

Since the mean squares are nonnegative, summing gives the interior pairing

$$\begin{aligned}
&-2\,\mathbb{E}\Big[|\delta X|^2 + |\delta Y|^2 + |\delta Z|_F^2 + \|\delta U\|_{L^2(\nu)}^2\Big]\\
&\quad - \Big(|\mathbb{E}\delta X|^2 + |\mathbb{E}\delta Y|^2 + |\mathbb{E}\delta Z|_F^2 + \|\mathbb{E}\delta U\|_{L^2(\nu)}^2\Big)\\
&\quad \le -2\,\mathbb{E}\Big[|\delta X|^2 + |\delta Y|^2 + |\delta Z|_F^2 + \|\delta U\|_{L^2(\nu)}^2\Big]
\end{aligned}$$

and the terminal pairing

$$\mathbb{E}\langle \delta g_T,\, \delta X_T\rangle = \mathbb{E}\langle 2\,\delta X_T + \mathbb{E}\,\delta X_T,\, \delta X_T\rangle = 2\,\mathbb{E}|\delta X_T|^2 + |\mathbb{E}\delta X_T|^2 \ge 2\,\mathbb{E}|\delta X_T|^2.$$

Assumption 2.5 therefore holds with $\alpha = \beta_1 = \beta_2 = 2$. The mean-field coupling costs nothing and supplies extra dissipation.

These are the coefficients of [14, Example 3.1], carried over here to an arbitrary σ-finite ν, including $\nu(E) = \infty$: the monotonicity hypothesis of their well-posedness theorem [14, (H3.2) and Theorem 3.1] bounds the total mass of their Lévy measure, whereas the verification above uses only $U \in L^2(\nu)$.

Example 5.2 (Brownian reduction). Remove the Poisson channel from the stochastic basis, so that the filtration is generated by W and $\mathcal{F}_0$ and the representation property is the Brownian one. Delete the U-coordinate from the tuple, reducing $\mathcal{H}$ to $\mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^{m\times d}$. Merely setting $h \equiv 0$ on a Brownian–Poisson basis is not a reduction, since the backward Poisson integral and its integrand U persist in the representation.

The assumptions reduce to W_2-Lipschitz joint-law dependence of (X, Y, Z) under G-monotonicity, the condition imposed in [3, (A1)] with rectangular full-rank G. With no jump channel the U-dissipativity constraint of Remark 2.10 is vacuous, so both rank branches of Assumption 2.5 are available.

Example 5.3 (Expectation-functional interactions). Let $\bar\phi : [0, T] \times \mathcal{H} \times \mathcal{H} \to \mathbb{R}^k$ be measurable and Lipschitz in (θ, θ') uniformly in t, and set

$$\phi(t, \theta, \mu) := \int_{\mathcal{H}} \bar\phi(t, \theta, \theta')\, \mu(\mathrm{d}\theta').$$

At each fixed t and θ this is the Bochner mean of the kernel $\bar\phi(t, \theta, \cdot)$, whose Lipschitz constant is at most $\mathrm{Lip}(\bar\phi)$, so (2.10) supplies the law-dependence component of Assumption 2.3. Its state-variable Lipschitz and integrability requirements are imposed on the full coefficient system.

This class contains the deterministic-coefficient, finite-activity independent-copy interactions of [14]. Their Lévy measure is the ν of the present setting. Let $\bar\psi$ be the two-point kernel of their drift, diffusion or driver, and $\bar h$ that of their jump coefficient. Both are assumed Lipschitz in all arguments except t, with constant uniform in t, and with square-integrable origin values, $\bar\psi(\cdot, 0, \dots, 0) \in L^2(0, T)$, the counterpart of the origin integrability imposed in [14, (H3.1)]. The pointwise-jump form embeds via

$$\bar\phi(t, \theta, \theta') = \int_E \bar\psi\Big(t, x, y, z, u(e), x', y', z', u'(e)\Big)\nu(\mathrm{d}e).$$

The integral is finite at every (θ, θ'): the Lipschitz bound on $\bar\psi$ and Cauchy–Schwarz in $L^2(\nu)$ give $\int_E \Big(|u(e)| + |u'(e)|\Big)\nu(\mathrm{d}e) \le \nu(E)^{1/2}\Big(\|u\|_{L^2(\nu)} + \|u'\|_{L^2(\nu)}\Big)$, and the origin term contributes $\nu(E)\, |\bar\psi(t, 0, \dots, 0)|$. The same bounds give, with a constant finite when $\nu(E) < \infty$,

$$\begin{aligned}&|\bar\phi(t, \theta_1, \theta_1') - \bar\phi(t, \theta_2, \theta_2')| \\ &\qquad \le \mathrm{Lip}(\bar\psi) \max\{\nu(E), \nu(E)^{1/2}\}\Big(|\theta_1 - \theta_2|_{\mathcal{H}} + |\theta_1' - \theta_2'|_{\mathcal{H}}\Big),\end{aligned}$$

so $\bar\phi$ is Lipschitz on $\mathcal{H} \times \mathcal{H}$. The pointwise values $u(e), u'(e)$ of the $L^2(\nu)$-components enter only under the ν-integral, so the functional is well defined on equivalence classes: changing u on a ν-null set changes the integrand on a ν-null set of marks.

In their jump coefficient the mark e is the integrator's and is not ν-integrated, so that coefficient embeds instead through the Nemytskii map

$$[\mathbf{H}_t(\theta,\theta')](e) := \bar{h}\Big(t,x,y,z,u(e),x',y',z',u'(e),e\Big) \quad \text{for } \nu\text{-a.e. } e.$$

This map is again well defined on equivalence classes. Assume it jointly Borel, and impose the natural quantitative hypotheses

$$\begin{gathered}\|\mathbf{H}_t(\theta_1,\theta_1') - \mathbf{H}_t(\theta_2,\theta_2')\|_{L^2(\nu)} \le C(|\theta_1-\theta_2|_{\mathcal{H}} + |\theta_1'-\theta_2'|_{\mathcal{H}}),\\ \int_0^T \|\mathbf{H}_t(0,0)\|^2_{L^2(\nu)}\,\mathrm{d}t < \infty,\end{gathered}$$

so that $\mathbf{H}_t$ is Lipschitz into $L^2(\nu;\mathbb{R}^n)$ and (2.10) applies, with that space in the role of K, to

$$h(t,\theta,\mu) := \int_{\mathcal{H}} \mathbf{H}_t(\theta,\theta')\,\mu(\mathrm{d}\theta').$$

Example 5.4 (Nonlinear law functionals). Fix

$$\begin{array}{lll} \mu_0 \in \mathcal{P}_2(\mathcal{H}), & \Lambda_0 \in \mathcal{P}_2(L^2(\nu)), & \varepsilon > 0,\\ \mathfrak{q},\tilde{\mathfrak{q}},\hat{\mathfrak{q}} : \mathbb{R}_+ \to \mathbb{R}, & Q : \mathbb{R}^J \to \mathbb{R}, & \varphi_1,\dots,\varphi_J : L^2(\nu) \to \mathbb{R},\\ w_0, w_1 \in \mathbb{R}^m, & v_0 \in \mathbb{R}^n, & \hat{h} \in L^2(\nu;\mathbb{R}^n), \end{array}$$

with $\mathfrak{q},\tilde{\mathfrak{q}},\hat{\mathfrak{q}},Q$ and the features φ_j bounded and Lipschitz, and $w_0,w_1,v_0,\hat{h}$ the perturbation directions.

Let the baseline generator (b_0,σ_0,h_0,f_0,g) be as in Corollary 3.6: $m=n$, so that the full-rank G is invertible; Assumptions 2.2–2.5 hold; and there is a dissipativity margin $c_{\mathrm{diss}} > 0$ in the unweighted form (2.13). Example 5.1(i) supplies such a baseline. Perturb by

$$\begin{aligned} f(t,\theta,\mu) &= f_0(t,\theta) + \varepsilon\,\mathfrak{q}\Big(W_2(\mu,\mu_0)\Big)w_0 + \varepsilon\,Q\left(\int_{L^2(\nu)}\varphi_1\,\mathrm{d}\mu^U,\dots,\int_{L^2(\nu)}\varphi_J\,\mathrm{d}\mu^U\right)w_1,\\ b(t,\theta,\mu) &= b_0(t,\theta) + \varepsilon\,\tilde{\mathfrak{q}}\Big(W_2(\mu^U,\Lambda_0)\Big)v_0,\\ h(t,\theta,e,\mu) &= h_0(t,\theta,e) + \varepsilon\,\hat{\mathfrak{q}}\Big(W_2(\mu^U,\Lambda_0)\Big)\hat{h}(e). \end{aligned}$$

The baseline is taken law-free for simplicity: a law-dependent dissipative baseline works identically.

The law functionals are of three types:

(i) $\mathfrak{q}(W_2(\mu,\mu_0))$, a nonlinear *radial* statistic of the full tuple law: a function of its W_2-distance to a fixed reference measure alone;

(ii) $\tilde{\mathfrak{q}}(W_2(\mu^U,\Lambda_0))$ and $\hat{\mathfrak{q}}(W_2(\mu^U,\Lambda_0))$, the same for the U-marginal, the latter entering the *jump coefficient itself*; and

(iii) the Q-term, a nonlinear *cylindrical* functional of the U-marginal: finitely many integrals of arbitrary bounded-Lipschitz features on the infinite-dimensional fibre $L^2(\nu)$, composed through the nonlinear Q.

All three lie beyond the expectation and two-point-kernel jump-integrand interactions of the preceding examples, and beyond the linear jump-integrand expectations of the cited models. All are W_2-Lipschitz in μ. The radial statistic has constant $\mathrm{Lip}(\mathfrak{q})$, by the triangle inequality for W_2. The U-marginal statistics have constants $\mathrm{Lip}(\tilde{\mathfrak{q}})$ and $\mathrm{Lip}(\hat{\mathfrak{q}})$, since marginal projection is 1-Lipschitz for W_2 by (2.11). Each cylindrical feature is the Bochner mean of the kernel $\theta' \mapsto \varphi_j(u')$, so (2.10) applies and the Q-term has constant

$$L_Q := \mathrm{Lip}(Q)\Big(\sum_{j=1}^{J} \mathrm{Lip}(\varphi_j)^2\Big)^{1/2}.$$

The perturbations are bounded multiples of the fixed vectors w_0, w_1, v_0 and of $\hat{h} \in L^2(\nu; \mathbb{R}^n)$, so they satisfy the origin-integrability condition of Assumption 2.2. They are independent of θ and satisfy the W_2-Lipschitz bound (3.5) with constant

$$L_p := \varepsilon\Big(|w_0|\,\mathrm{Lip}(\mathfrak{q}) + |v_0|\,\mathrm{Lip}(\tilde{\mathfrak{q}}) + |w_1|\,L_Q + \|\hat{h}\|_{L^2(\nu)}\,\mathrm{Lip}(\hat{\mathfrak{q}})\Big),$$

hence the diagonal Lipschitz condition of Assumption 2.3 as well.

By (3.5) and Corollary 3.6, (3.4) holds with $c_{\mathrm{pert}} = \|G\| L_p$, so the perturbed system is well-posed, with margin $c_{\mathrm{diss}} - \|G\| L_p$, whenever

$$\varepsilon\,\|G\|\Big(|w_0|\,\mathrm{Lip}(\mathfrak{q}) + |v_0|\,\mathrm{Lip}(\tilde{\mathfrak{q}}) + |w_1|\,L_Q + \|\hat{h}\|_{L^2(\nu)}\,\mathrm{Lip}(\hat{\mathfrak{q}})\Big) \;<\; c_{\mathrm{diss}}. \tag{5.1}$$

The smallness in (5.1) binds the law-dependence increment alone; the forward–backward coupling of the baseline is unrestricted, the opposite trade to the weak-coupling route surveyed in the introduction.

Example 5.5 (A non-perturbative full U-law model: the mean-field dealer market). Read the control problem of Example 5.1(i) as a dealer managing a one-factor marked-to-market exposure X: requests for quotes arrive as the marked points of N, the mark e recording side and size, and a fill of mark e moves the exposure through the jump channel. The Brownian term is the dealer's residual hedging risk; marking to market removes the common reference price from the strategic problem. The adjoint component U_t is then the dealer's *valuation curve*: $U_t(e)$ is the marginal continuation value of a fill of mark e, so U_t is an element of $L^2(\nu)$ indexed by side and size, and a population of dealers carries a law on curves, $\mu^U \in \mathcal{P}_2(L^2(\nu))$. Quotes are affine functions of this curve, so the population's quote-curve distribution is an affine image of μ^U, and an anonymous routing mechanism that compares each dealer's curve with the population's tilts the filled flow by the *crowding correction*

$$\mathcal{A}_\lambda(u) := \int_{L^2(\nu)} \nabla V(u - v)\,\lambda(\mathrm{d}v), \qquad V(w) := \sqrt{1 + \|w\|^2_{L^2(\nu)}} - 1, \qquad \lambda \in \mathcal{P}_2(L^2(\nu)).$$

The potential V is even and convex, with

$$\nabla V(w) = \frac{w}{\sqrt{1+\|w\|^2_{L^2(\nu)}}}, \qquad \|\nabla V(w)\|_{L^2(\nu)} \le 1, \qquad \mathrm{Lip}(\nabla V) \le 1,$$

the last because the second derivative of V has operator norm at most one. Fix $\varepsilon \ge 0$ and tilt the jump response of Example 5.1(i) by the crowding correction:

$$b = -\beta_1 y, \qquad \sigma = \sigma_c - \beta_1 z, \qquad h = h_c - \beta_1 u - \varepsilon\,\mathcal{A}_{\mu^U}(u), \qquad f = \beta_2 x, \qquad g(x,\rho) = x,$$

with μ^U the U-marginal of the measure argument. At $\varepsilon = 0$ this is Example 5.1(i) verbatim.

Assumptions 2.2–2.4. Since $\nabla V(0) = 0$, the origin value $h(\cdot, 0, \delta_0) = h_c$ is unchanged, so Assumption 2.2 holds as in Example 5.1(i). For $u, u' \in L^2(\nu)$ and $\lambda, \lambda' \in \mathcal{P}_2(L^2(\nu))$, coupling (v, v') optimally for (λ, λ') and using that ∇V is 1-Lipschitz,

$$\|\mathcal{A}_\lambda(u) - \mathcal{A}_{\lambda'}(u')\|_{L^2(\nu)} \;\le\; \|u-u'\|_{L^2(\nu)} + W_2(\lambda, \lambda'),$$

and $W_2(\mu^U, \mu'^U) \le W_2(\mu, \mu')$ by (2.11), so Assumption 2.3 holds with $L = \max\{\beta_1 + \varepsilon, \beta_2, 1\}$: the interaction strength enlarges the Lipschitz constant only. Joint Lipschitz continuity of $(u, \lambda) \mapsto \mathcal{A}_\lambda(u)$ gives the Borel measurability (2.3). Assumption 2.4 holds as in Example 5.1(i).

Monotonicity at arbitrary ε. Let $(\Theta, \mu), (\Theta', \mu') \in \mathcal{D}_t$; taking U-marginals of $\mu = \mathrm{Law}(\Theta)$ and $\mu' = \mathrm{Law}(\Theta')$, $\mu^U = \mathrm{Law}(U)$ and $\mu'^U = \mathrm{Law}(U')$. Let $(\widehat{U}, \widehat{U}')$ be an independent copy of (U, U'): then $\widehat{U}$ has law μ^U and is independent of (U, U'), so

$$\mathcal{A}_{\mu^U}(U) = \int_{L^2(\nu)} \nabla V(U - v)\,\mu^U(\mathrm{d}v) = \mathbb{E}\Big[\nabla V(U - \widehat{U})\,\Big|\,U\Big],$$

and likewise for (U', μ'^U). Since exchanging (U, U') with $(\widehat{U}, \widehat{U}')$ preserves the joint law while ∇V is odd,

$$\begin{aligned} \mathbb{E}\langle \mathcal{A}_{\mu^U}(U) - \mathcal{A}_{\mu'^U}(U'),\, U - U'\rangle \;&=\; \mathbb{E}\langle \nabla V(U - \widehat{U}) - \nabla V(U' - \widehat{U}'),\, U - U'\rangle \\ &=\; -\,\mathbb{E}\langle \nabla V(U - \widehat{U}) - \nabla V(U' - \widehat{U}'),\, \widehat{U} - \widehat{U}'\rangle. \end{aligned}$$

Averaging the two lines,

$$\begin{aligned} &\mathbb{E}\langle \mathcal{A}_{\mu^U}(U) - \mathcal{A}_{\mu'^U}(U'),\, U - U'\rangle \\ &\quad = \tfrac{1}{2}\,\mathbb{E}\langle \nabla V(U - \widehat{U}) - \nabla V(U' - \widehat{U}'),\, (U - \widehat{U}) - (U' - \widehat{U}')\rangle \;\ge\; 0, \end{aligned}$$

the final inequality because ∇V, the gradient of the convex V, is monotone. The ε-term therefore improves the interior pairing,

$$\mathbb{E}\int_E \langle \delta h(e),\, \delta U(e)\rangle\,\nu(\mathrm{d}e) \;\le\; -\beta_1\,\mathbb{E}\|\delta U\|^2_{L^2(\nu)},$$

and (2.7) holds with the constants $\alpha = 1, \beta_1, \beta_2$ of Example 5.1(i) for *every* $\varepsilon \geq 0$ — no smallness condition and no margin. Theorem 3.5 applies directly.

Beyond the mean. The crowding correction is not a function of $\mathbb{E}[U]$: for a unit $w \in L^2(\nu)$, the two-point laws $\lambda_1 = \frac{1}{2}(\delta_{-w} + \delta_w)$ and $\lambda_2 = \frac{1}{2}(\delta_{-2w} + \delta_{2w})$ share their mean, yet

$$\mathcal{A}_{\lambda_1}(w) = \tfrac{1}{\sqrt{5}}\, w, \qquad \mathcal{A}_{\lambda_2}(w) = \tfrac{1}{2}\Big(\tfrac{3}{\sqrt{10}} - \tfrac{1}{\sqrt{2}}\Big) w.$$

Marks and activity. Take $E = \{-1, +1\} \times (0, 1]$, writing $e = (\varsigma, q)$ for side ς and size q, with $\nu(\mathrm{d}\varsigma, \mathrm{d}q) = c_\varsigma\, q^{-1-\varrho}\, \mathrm{d}q$ for constants $c_\varsigma > 0$ and $0 < \varrho < 2$: then $\nu(E) = \infty$ while $\int_E q^2\, \nu(\mathrm{d}\varsigma, \mathrm{d}q) < \infty$, so $h_c(\varsigma, q) := \varsigma q$ lies in $L^2(\nu)$ — infinitely many small tickets with square-integrable aggregate exposure. In this model the infinite-activity regime is the market's small-ticket limit.

The finite-player inspiration is the dealer market of Cont and Xiong [8], in which dealers' execution probabilities depend on their own and competing quotes; the present example is a stylised mean-field model inspired by that setting.

Remark 5.6 (Scope of the measure argument). The proofs consume the tuple law through a single mechanism: the W_2 term of Assumption 2.3, dominated at every use by the synchronous coupling bound (2.8). The right-hand side of (2.8) carries $\mathbb{E}\|\delta U\|^2_{L^2(\nu)}$ whether or not the measure argument reads the U-marginal, because the coefficients depend on u pointwise — a dependence Remark 2.10 makes unavoidable whenever $\beta_1 > 0$. Nothing is therefore gained by restricting the coefficients to the (X, Y, Z)-marginal law: with the coupling bound taken in $\mathcal{H}$, the estimates close at the full tuple law, and Example 5.5 exhibits that scope arising from a routing mechanism rather than by construction.

Remark 5.7 (Perturbative and non-perturbative law dependence). Example 5.4 shows that arbitrary nonlinear full-tuple law dependence can be added when it is smaller than the dissipativity margin. Example 5.5 shows that a genuinely nonlinear U-law interaction can have arbitrary strength when its geometry is itself monotone.

6 Conclusion

Theorems 3.1, 3.3 and 3.5 establish global well-posedness, with a quantitative stability estimate, for fully coupled MV-FBSDEJ with W_2-dependence on the law of the whole tuple, including the $L^2(\nu)$-valued jump integrand, at arbitrary σ-finite jump activity. By Corollary 3.6, the well-posedness is robust, at $m = n$, under full-tuple-law perturbations below the dissipativity margin. The Lipschitz and monotonicity assumptions are imposed only along the diagonal inputs $(\Theta_t, \mathrm{Law}(\Theta_t)) \in \mathcal{D}_t$: Proposition 3.8 shows that expected diagonal G-monotonicity is strictly weaker than its pointwise form (Remark 2.8). The monotonicity of Assumption 2.5 is the principal structural assumption and is still coercive; in the $m < n$ regime it forces strictly U-dissipative jump feedback (Remark 2.10). Relaxing that coercivity, so that additive jump coefficients become admissible when $m < n$, and treating domination-monotonicity conditions [21, 23] or conditional laws under common noise, are the natural next questions.

A Predictable representatives and the law flow

Lemma A.1 (Predictable representatives and the full-tuple law flow)**.** *Let X, Y be càdlàg adapted and let Z, U be fixed predictable versions, with*

$$\mathbb{E}\int_0^T \Big(|X_t|^2 + |Y_t|^2 + |Z_t|_F^2 + \int_E |U_t(e)|^2\,\nu(\mathrm{d}e)\Big)\mathrm{d}t < \infty.$$

Then the a.e.-defined maps $t \mapsto \mathrm{Law}(\Theta_t^-)$ and $t \mapsto \mathrm{Law}(\Theta_t)$ admit Borel representatives, unique up to Lebesgue-null sets of times,

$$\mu^-,\, \mu : [0,T] \to (\mathcal{P}_2(\mathcal{H}), W_2), \qquad \mu_t^- = \mu_t \quad \textit{for a.e. } t.$$

Coefficient evaluations along these representatives are invariant, up to $\mathrm{d}t\otimes\mathrm{d}\mathbb{P}$-null sets, under the choice of predictable representatives of (Z,U) and of the Borel representatives.

Proof. Step 1 (the field). The left limits X_-, Y_- are predictable and agree with X, Y $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$-a.e., as càdlàg paths jump at countably many times, so $\int_0^T \mathbf{1}_{\{X_{t-}\neq X_t\}}\,\mathrm{d}t = 0$ almost surely; the hypothesis on X, Y thus passes to them, and $\mathcal{H}$ is separable because $L^2(\nu)$ is. The Z-coordinate is predictable and square-integrable by assumption. For the U-coordinate, the integrability hypothesis and the Fubini isometry (2.4) give

$$U \in L^2\Big([0,T]\times\Omega\times E\Big) \cong L^2\Big([0,T]\times\Omega; L^2(\nu)\Big),$$

so the induced map $(t,\omega) \mapsto U_t(\omega,\cdot)$ admits a predictable, strongly measurable $L^2(\nu)$-valued representative. Θ^- is therefore a predictable, hence jointly measurable, $\mathcal{H}$-valued field with

$$\mathbb{E}\int_0^T \|\Theta_t^-\|_{\mathcal{H}}^2\,\mathrm{d}t < \infty, \qquad \text{that is,} \qquad \Theta^- \in L^2\Big([0,T]\times\Omega;\mathcal{H}\Big).$$

The Fubini isometry of [11, Proposition 1.2.24],

$$L^2\Big([0,T]\times\Omega;\mathcal{H}\Big) \;\cong\; L^2\Big(0,T; L^2(\Omega;\mathcal{H})\Big),$$

supplies a strongly measurable $L^2(\Omega;\mathcal{H})$-valued representative of $t \mapsto \Theta_t^-$; by the Pettis measurability theorem [11, Theorem 1.1.6], that map takes values in a separable closed subspace of $L^2(\Omega;\mathcal{H})$ off a Lebesgue-null set of times, and therefore admits a Borel version, agreeing with it for a.e. t.

Step 2 (Borel flow). By the synchronous coupling bound (2.8), the law map $\Theta \mapsto \mathrm{Law}(\Theta)$ is 1-Lipschitz from $L^2(\Omega;\mathcal{H})$ into $(\mathcal{P}_2(\mathcal{H}), W_2)$, which inherits Polishness from $\mathcal{H}$ [22, Ch. 6]. Composing it with the Borel version of Step 1 gives a Borel map $\mu^- : [0,T] \to \mathcal{P}_2(\mathcal{H})$ with $\mu_t^- = \mathrm{Law}(\Theta_t^-)$ for a.e. t; any two such representatives agree a.e. The same argument applied to the progressively measurable field Θ yields μ; as Θ and Θ^-

agree $dt \otimes d\mathbb{P}$-a.e., $\mu_t = \mu_t^-$ for a.e. t.

Step 3 (independence of representatives). Let (Z', U') be another pair of predictable versions, so $Z = Z'$ $dt \otimes d\mathbb{P}$-a.e. and $U = U'$ $dt \otimes d\mathbb{P} \otimes \nu$-a.e.; by Fubini the latter gives $U = U'$ in $L^2(\nu)$ for $dt \otimes d\mathbb{P}$-a.e. (t, ω). The two fields therefore agree in $L^2(\Omega; \mathcal{H})$ for a.e. t, whence $\mu_t^- = (\mu_t^-)'$ for a.e. t. The coefficients are therefore evaluated at arguments coinciding $dt \otimes d\mathbb{P}$-a.e. (in $L^2(\nu)$ for the jump coefficient); changing the Borel representative likewise alters μ^- only on a Lebesgue-null set of times, hence the arguments only on a $dt \otimes d\mathbb{P}$-null set. In either case the coefficient processes represent the same L^2 classes over their respective product measures, so every Lebesgue and stochastic integral built from them, hence every identity and estimate, is unchanged. □

References

[1] D. Applebaum. *Lévy Processes and Stochastic Calculus.* 2nd ed., Cambridge Studies in Advanced Mathematics 116, Cambridge University Press, Cambridge, 2009; doi:10.1017/CBO9780511809781.

[2] A. Bensoussan, Z. Huang, S. Tang, S. C. P. Yam. *Mean field type control problems driven by jump-diffusions.* arXiv:2509.25614, 2025. Preprint.

[3] A. Bensoussan, S. C. P. Yam, Z. Zhang. *Well-posedness of mean-field type forward–backward stochastic differential equations.* Stochastic Process. Appl. 125(9):3327–3354, 2015; doi:10.1016/j.spa.2015.04.006.

[4] R. Carmona, F. Delarue. *Mean field forward–backward stochastic differential equations.* Electron. Commun. Probab. 18(68):1–15, 2013; arXiv:1211.4186; doi:10.1214/ECP.v18-2446.

[5] T. Chen, H. Shi, Z. Wu. *A progressive maximum principle of fully coupled mean-field system with jumps.* J. Optim. Theory Appl. 206(3), Article 75, 2025; doi:10.1007/s10957-025-02760-y.

[6] Y. Chen, T. Nie. *Well-posedness of fully coupled McKean–Vlasov FBSDE and application to Stackelberg games.* ESAIM Control Optim. Calc. Var. 31 (2025), Paper No. 56; doi:10.1051/cocv/2025041.

[7] Y. Chen, T. Nie, S. Wang. *Fully-coupled mean-field FBSDE and maximum principle for related optimal control problem.* Systems Control Lett. 177:105550, 2023; doi:10.1016/j.sysconle.2023.105550.

[8] R. Cont, W. Xiong. *Dynamics of market making algorithms in dealer markets: Learning and tacit collusion.* Math. Finance 34(2):467–521, 2024; doi:10.1111/mafi.12401.

[9] S.-W. He, J.-G. Wang, J.-A. Yan. *Semimartingale Theory and Stochastic Calculus.* Science Press, Beijing; CRC Press, Boca Raton, 1992.

[10] T. Hua, P. Luo. *Well-posedness for a class of mean-field-type forward–backward stochastic differential equations and classical solutions of related master equations.* J. Theoret. Probab. 38 (2025), Paper No. 12; arXiv:2307.11536; doi:10.1007/s10959-024-01375-9.

[11] T. Hytönen, J. van Neerven, M. Veraar, L. Weis. *Analysis in Banach Spaces. Volume I: Martingales and Littlewood–Paley Theory.* Ergebnisse der Mathematik und ihrer Grenzgebiete 63, Springer, Cham, 2016; doi:10.1007/978-3-319-48520-1.

[12] O. Kallenberg. *Foundations of Modern Probability.* 3rd ed., Probability Theory and Stochastic Modelling 99, Springer, Cham, 2021; doi:10.1007/978-3-030-61871-1.

[13] J. Li. *Mean-field forward and backward SDEs with jumps and associated nonlocal quasi-linear integral-PDEs.* Stochastic Process. Appl. 128(9):3118–3180, 2018; arXiv:1702.05213; doi:10.1016/j.spa.2017.10.011.

[14] W. Li, H. Min. *Fully coupled mean-field FBSDEs with jumps and related optimal control problems.* Optim. Control Appl. Methods 42(1):305–329, 2021; arXiv:1812.10254; doi:10.1002/oca.2677.

[15] J. Liu, J. Zhang. *A probabilistic interpretation of the master equation arising from mean field games with jump diffusion.* arXiv:2601.19084, 2026. Preprint.

[16] A. Matoussi, A. Manai, R. Salhi. *Mean-field backward-forward SDE with jumps and storage problem in smart grids.* arXiv:1906.08525, 2019. Preprint.

[17] H. Min, Y. Peng, Y. Qin. *Fully coupled mean-field forward–backward stochastic differential equations and stochastic maximum principle.* Abstr. Appl. Anal. 2014:839467, 2014; doi:10.1155/2014/839467.

[18] S. Peng, Z. Wu. *Fully coupled forward–backward SDEs and applications to optimal control.* SIAM J. Control Optim. 37(3):825–843, 1999; doi:10.1137/S0363012996313549.

[19] C. Reisinger, W. Stockinger, Y. Zhang. *A posteriori error estimates for fully coupled McKean–Vlasov forward-backward SDEs.* IMA J. Numer. Anal. 44:2323–2369, 2024; doi:10.1093/imanum/drad060.

[20] S. Tang, X. Li. *Necessary conditions for optimal control of stochastic systems with random jumps.* SIAM J. Control Optim. 32(5):1447–1475, 1994; doi:10.1137/S0363012992233858.

[21] R. Tian, Z. Yu. *Mean-field type FBSDEs under domination-monotonicity conditions and application to LQ problems.* SIAM J. Control Optim. 61(1):22–46, 2023; arXiv:2203.14654; doi:10.1137/21M144219X.

[22] C. Villani. *Optimal Transport: Old and New.* Grundlehren der mathematischen Wissenschaften 338, Springer, Berlin, 2009; doi:10.1007/978-3-540-71050-9.

[23] H. Wu. *Extended MF-FBSDEs with nonlinear domination-monotonicity conditions and stochastic optimal controls of linear system with quadruple controls.* arXiv:2605.09374, 2026. Preprint.

[24] Z. Wu. *Forward–backward stochastic differential equations with Brownian motion and Poisson process.* Acta Math. Appl. Sin. (Engl. Ser.) 15(4):433–443, 1999; doi:10.1007/BF02684045.

[25] M. Xu, M. Tang, Q. Meng. *A class of forward–backward stochastic differential equations driven by Lévy processes and application to LQ problems.* Chinese Ann. Math. Ser. A 46(4):403–434, 2025 (in Chinese); arXiv:2310.13195; doi:10.16205/j.cnki.cama.2025.0026.

[26] X. Zhang, Z. Huang. *Well-posedness and comparison theorems of coupled mean-field forward–backward stochastic differential equations.* Systems Control Lett. 209:106351, 2026; doi:10.1016/j.sysconle.2026.106351.